CONGRUENT DISSECTIONS OF A TRIANGLE: PRIME COUNTS, THE COUNTS UP TO 200, AND A FAMILY OF ISOSCELES TARGETS

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ABSTRACT. A positive integer N is a *tile count* if some triangle can be cut into N pairwise congruent triangles; Erdős asked for the set of tile counts. The problem is not solved here. The results below are theorems of an accompanying Lean 4 library unless stated otherwise. (1) A prime p is a tile count if and only if $p = 3$ or $p \not\equiv 3 \pmod 4$. This is a theorem of the formal library, with the standard axioms; the formal statement has not yet been compared with the mathematical one by an independent reader. (2) For $1 \le N \le 200$ the tile counts are determined; the largest search certificate, which excludes 135, has 1 326 256 nodes and is checked by the Lean kernel. (3) For a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational, a signed-direction functional on directed edges gives two integrality conditions; no prime number of such tiles tiles an isosceles, non-equilateral triangle. (4) For coprime $1 \le e < f$ let $T_M(e, f)$ be the isosceles triangle with legs Mf^3 and base $Me(3f^2 - e^2)$, with the tile $(ef, f^2 - e^2, f^2)$. A tiling exists for every $M \ge f$ (constructions of D. Paliy and M. Beeson, formalized here); there is none for $M = 1$, none when $2Me < f$, and for $e = 1$ none when $2M \le f$. The scales $M \ge 2$ with $f/(2e) < M < f$ are open, apart from three instances; the open part is stated precisely.

1. INTRODUCTION

1.1. **The problem.** Let T be a triangle. A triangle Δ is *cut into N copies of T*, or *tiled by N copies of T*, if Δ is the union of N closed triangles with pairwise disjoint interiors, each congruent to T. Mirror images of T are allowed, and the pieces need not meet edge to edge: a vertex of one piece may lie in the relative interior of a side of another. We call Δ the *target* and T the *tile*, and we say that N is a *tile count* if some triangle can be cut into N copies of some triangle. Erdős asked for the set of tile counts; see Soifer [23]. It is Problem 634 of the collection at `erdosproblems.com`, where $N = 19$ is listed as an open instance.

Every square is a tile count, for any triangle, and so are the numbers $x^2 + y^2$, $3k^2$ and $6k^2$ [23]. Laczkovich [11, 12] determined which pairs of shapes (Δ, T) can occur, and Beeson examined the resulting classes one at a time [1, 3, 4, 5, 6]. Zhang [25] constructed families of tilings by tiles with an angle $2\pi/3$, and Beeson, Laczkovich and Zhang settled the neighbouring Problem 633 [8]. Among the primes, 2, 3 and the primes $\equiv 1 \pmod 4$ are tile counts by the classical constructions. The primes $p \equiv 3 \pmod 4$ with $p > 3$ are the odd primes that are neither sums of two squares nor of the form $3k^2$, and it is a folklore conjecture that none of them is a tile count. Beeson proved this for $p = 7$ [5].

The paper is accompanied by a library for the proof assistant Lean 4 with Mathlib, in the same repository; Appendix A describes it and maps each numbered statement to its Lean declaration. Problem 634 is not solved here. Section 8 states what a solution still requires.

1.2. **Results.** One family of pairs (Δ, T) needs notation from the start (Definition 2.6). For coprime integers $1 \le e < f$ the triangle with sides $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$ has angles $(\alpha, \beta, 2\alpha + \beta)$ with $3\alpha + 2\beta = \pi$ and α/π irrational. For an integer $M \ge 1$ let $T_M(e, f)$ be the isosceles triangle with legs Mf^3 and base $Me(3f^2 - e^2)$; its base angles are β and its apex is 3α. Let $W_M(e, f)$ be the triangle with sides $Mf(f^2 - e^2)$, $Me(2f^2 - e^2)$, Mf^3; its angles are β,

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2α, $\alpha+\beta$. Their areas are $M^2(3f^2-e^2)$ and $M^2(2f^2-e^2)$ times the area of the tile. We call (e,f) a *member*, M the *scale*, and the triangles $T_M(e,f)$ the *base-β targets*.

Each numbered theorem, proposition, lemma, corollary and computation of this paper carries a mark F, W or S, explained in Section 1.4: F for a theorem of the Lean library, W for a written proof that is not formalized, S for the outcome of a computer search without a certificate.

Theorem 1.1 (Prime counts; F)**.** *Let p be a prime. Some triangle can be cut into p congruent triangles if and only if $p=3$ or $p\not\equiv 3 \pmod 4$.*

Standing of Theorem 1.1. The theorem is formalized in Lean 4 with Mathlib and depends on the standard axioms only. The formal statement and the definitions it uses have been compared with the mathematical ones within this work, and not yet by an independent reader. With this qualification, it says that no prime $p\equiv 3 \pmod 4$ with $p>3$ is a tile count, and in particular that no triangle can be cut into 19 congruent triangles. The formal statement is displayed in Section 4.4.

Priority. A proof of Theorem 1.1 was first made public by D. Paliy on 29 September 2026 [18], by a column induction from a corner of the scale-one targets. The proof given here was obtained independently, by different arguments at scale one (Section 4.3); its exclusion of the family $e=1$ was public on 12 September 2026 [19], and the rest was completed without being made public at the time.

The formal proof of Theorem 1.1 is a case analysis on the tile. Arithmetic conditions close every case except three targets of the tile (ef, f^2-e^2, f^2): the base-β target $T_1(e,f)$, with $3f^2-e^2$ tiles, the target $W_1(e,f)$, with $2f^2-e^2$ tiles, and one isosceles target. The last two lead back to the first (Section 4.2). The primes of the form $3f^2-e^2$ are the primes $\equiv 11 \pmod{12}$ (Proposition 4.1). For these three targets the theorem rests on the following statement, which concerns every member and any number of tiles.

Theorem 1.2 (The base-β targets at scale one; F)**.** *Let $1\le e<f$ be coprime integers. No triangle with sides f^3, f^3, $e(3f^2-e^2)$ can be cut into congruent triangles with sides ef, f^2-e^2, f^2, in any number.*

The standing of Theorem 1.2 is that of Theorem 1.1.

Theorem 1.3 (Tile counts up to 200; F)**.** *Let $1\le N\le 200$. Then N is a tile count if and only if N is one of the 105 numbers of Table 1. These are the numbers $N\le 200$ of the forms*

$$k^2, \qquad 3k^2, \qquad 6k^2, \qquad x^2+y^2 \ \ (x,y\ge 1), \qquad 7m^2, \ \ 11m^2 \ \ (m\ge 2),$$

together with 77, 84, 88, 126, 143 and 189.

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Theorem 1.4 (Isosceles targets of a tile with an angle $2\pi/3$; F). *Let T have angles $(\alpha, \beta, 2\pi/3)$ with α/π irrational. Then no prime number of copies of T tiles an isosceles triangle that is not equilateral.*

For this target Beeson [4] showed that a tiling needs at least 36 tiles and left the prime case open. The statement of Theorem 1.4 is also proved by Beeson, by a different argument [2, Thms. 8 and 10]; the proof given here uses the signed-direction functional of Section 3.

Theorem 1.5 (Tilings at every scale $M \geq f$; D. Paliy, M. Beeson; F). *Let $1 \leq e < f$ be coprime integers and $M \geq f$. Then $W_M(e, f)$ and $T_M(e, f)$ can be cut into congruent triangles with sides $(ef,\ f^2 - e^2,\ f^2)$.*

The statement and the constructions are not ours. The statement is announced by Paliy [20]; the tiling at $M = f$ is a construction that his notes credit to Beeson [3, Thm. 11], and the passage to all $M \geq f$ is by his exterior collar and his additional seeds. What is added here is a formal proof for all members and scales.

Theorem 1.6 (Runs on a side, members $(1, f)$; F). *Let $f \geq 2$ be an integer and $(a, b, c) = (f,\ f^2 - 1,\ f^2)$, with angles α, β, γ opposite a, b, c. Let Δ be a triangle with a side s whose two corners have the angles β, β or $\alpha + \beta, \beta$, and suppose that Δ is cut into copies of the triangle (a, b, c). Then among the tiles with an edge on s there are f consecutive ones that lay their side c on s and show the angle α at the same end of it. These f tiles are translates of one another, and $|s| \geq f^3$.*

Corollary 1.7 (Members $(1, f)$ below the scale f; F). *Let $f \geq 2$ and $M \geq 1$ be integers.*

(a) *$W_M(1, f)$ can be cut into copies of $(f, f^2 - 1, f^2)$ if and only if $M \geq f$.*
(b) *If $2M \leq f$, then $T_M(1, f)$ cannot be cut into copies of $(f, f^2 - 1, f^2)$.*

The next three statements extend Theorem 1.6 and part of Corollary 1.7 from the members $(1, f)$ to every member. All three are formal.

Theorem 1.8 (Runs on a side, every member; F). *Let $1 \leq e < f$ be coprime integers and $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$, with angles α, β, γ opposite a, b, c. Let Δ be a triangle with a side s whose two corners have the angles β, β or $\alpha + \beta, \beta$, and suppose that Δ is cut into copies of the triangle (a, b, c). Then among the tiles with an edge on s there are f consecutive ones that lay their side c on s and show the angle α at the same end of it. In particular $|s| \geq f^3$.*

Corollary 1.9 (Every member below the scale f: two sides; F). *Let $1 \leq e < f$ be coprime integers, $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$, and $M \geq 1$.*

(a) *If $Me < f$, then $W_M(e, f)$ cannot be cut into copies of (a, b, c).*
(b) *If $2Me < f$, then $T_M(e, f)$ cannot be cut into copies of (a, b, c).*

Corollary 1.10 (Every member below the scale f: the third side of $W_M(e, f)$; F). *Let $1 \leq e < f$ be coprime integers, $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$, and $M \geq 1$.*

(a) *If $M(f - e) \leq e$, then $W_M(e, f)$ cannot be cut into copies of (a, b, c).*
(b) *For every $f \geq 2$, the triangle $W_M(f-1, f)$ can be cut into copies of $(f(f-1),\ 2f-1,\ f^2)$ if and only if $M \geq f$.*
(c) *The triangle $W_2(3, 5)$, with sides 160, 246, 250, cannot be cut into copies of $(15, 16, 25)$.*

In part (b) of Corollary 1.10 the tilings for $M \geq f$ are those of Theorem 1.5; the exclusion for $M < f$ comes from part (a). Both halves are formal. For $e \geq 2$ the inequality $2Me \leq f$ is the same as $2Me < f$, since e and f are coprime; so part (b) of Corollary 1.9 is the counterpart of Corollary 1.7(b).

Proposition 1.11 (A layer of $T_M(1,f)$; F)**.** *Let $f \geq 2$ be an integer.*

(a) *If $M \geq 1$, $2M \geq f$, and $T_M(1,f)$ has a tiling by the tile (f, f^2-1, f^2), so has $T_{M+1}(1,f)$.*
(b) *There is an integer $M^*(f)$ with $f/2 < M^*(f) \leq f$ such that, for $M \geq 1$, $T_M(1,f)$ has a tiling if and only if $M \geq M^*(f)$.*
(c) *$M^*(f) = f$, that is, $\tau(1,f,M)$ holds for every $M \geq 1$, if and only if $T_{f-1}(1,f)$ has no tiling.*

The tiling of (a) *is explicit: for $2M' \geq f+2$ the trapezoid cut from $T_{M'}(1,f)$ by the line parallel to the base at the height H_1 of $T_1(1,f)$ is cut into $(2M'-1)(3f^2-1)$ copies of the tile (the* layer*), and the part above it is a translate of $T_{M'-1}(1,f)$.*

Proposition 1.11 is a theorem of the Lean library in the form (a)–(c); the description of the layer is the construction of its proof.

In every case decided so far $T_M(e,f)$ has a tiling exactly when $M \geq f$. Whether this is the rule is open:

Question 1.12 (The scales below f)**.** Let $1 \leq e < f$ be coprime and $2 \leq M < f$. Can $T_M(e,f)$ be cut into copies of (ef, f^2-e^2, f^2)?

The answer is no for $(e,f,M) = (2,3,2)$ and $(3,4,2)$, for $e=1$ and $2M \leq f$, and for $(1,3,2)$ (Proposition 7.2, Corollary 1.7); these answers are formal. By Corollary 1.9(b), which is formal as well, the answer is no whenever $2Me < f$; for $e \geq 2$ and $M \geq 2$ this requires $f > 4e$, and the first such case is $T_2(2,9)$. The open cases are therefore those with $f/(2e) < M < f$ other than $(1,3,2)$, $(2,3,2)$ and $(3,4,2)$. For $e=1$ these are $f/2 < M < f$ with $f \geq 4$, and by Proposition 1.11 they are governed by the single case $M = f-1$: if $T_{f-1}(1,f)$ has no tiling, neither has $T_M(1,f)$ for any $M < f$. The first such cases are $T_3(1,4)$, with 423 tiles, and $T_4(1,5)$, with 1184. For $e \geq 2$ the first open cases are $T_3(3,4)$, with 351 tiles, and the scales $2 \leq M \leq 4$ of the members $(2,5)$, $(3,5)$, $(4,5)$; for a member with $2e > f$ every scale $2 \leq M < f$ is open, apart from $(2,3,2)$ and $(3,4,2)$. For $e \geq 2$ a layer exists as well, of e times the height of $T_1(e,f)$ (Proposition 7.4, formal); for the members $(f-1,f)$ it exists only at the scales $M \geq f$ and does not reduce the open scales. We do not know the answer in any open case.

For the triangles $W_M(e,f)$ more is known. They are settled for the members $(1,f)$ (Corollary 1.7) and for the members $(f-1,f)$ (Corollary 1.10), both formal: a tiling exists exactly when $M \geq f$. For the other members there is none when $Me < f$ or $M(f-e) \leq e$, and the scales $M < f$ outside these two ranges are open, apart from $W_2(3,5)$; the first are $W_3(2,5)$, $W_4(2,5)$, $W_3(3,5)$ and $W_4(3,5)$. By Proposition 7.5, which is formal, the open scales of a member are governed by the single scale $M = f-1$: if $W_{f-1}(e,f)$ has no tiling, neither has $W_M(e,f)$ for any $M < f$.

Further results are stated in the sections: the commensurability of the sides of a tile with an angle $\pi/3$ or $2\pi/3$ that tiles an equilateral triangle, which are theorems of Laczkovich and of Beeson and Zhang formalized here (Theorem 5.2); statements for infinitely many composite N (Section 6.4); and the members for which Question 1.12 is answered at every scale (Proposition 7.6).

1.3. **Ideas of the proofs.** *Theorem 1.4.* For a tile with an angle $2\pi/3$ one would like to colour the tiles in two colours and count. This is not available, because a vertex of a tiling may be surrounded by an even number of tiles. We weight directed edges instead. Every edge of a tiling has a direction $j\frac{\pi}{3} + k\alpha$ with integers j,k, and $(-1)^j$ depends only on the direction because α/π is irrational; the weight of a directed edge is its length times $(-1)^j$. The weight changes sign when the edge is reversed, so the two sides of an interior segment cancel, and it is linear in the length, so the cancellation does not depend on how the segment is subdivided on either

side. Around a tile the weights add up to $\pm(c+a-b)$. The move exchanges a sum over the N tiles, about which nothing is known, for the weight of the boundary of the target, which is explicit: the latter is an integer multiple of $c+a-b$, and the integer has the parity of N. For an isosceles target and prime N the integer would be $(c-a-b)/\sqrt{b}$, with a primitive integer triple $c^2=a^2+ab+b^2$, and this is never an integer.

Theorem 1.1. The tile is sorted into finitely many types by its angles (Proposition 2.5), and for each type the area and the two boundary weights give equations between integers. For a prime $N \equiv 3 \pmod 4$, $N>3$, these equations have no solution in any case but one. In the remaining case the tile is (ef, f^2-e^2, f^2), and the target is $T_1(e,f)$, or it is $W_1(e,f)$ or an isosceles triangle with base angles $\alpha+\beta$, and in the last two cases gluing one or two scaled tiles to the target produces a tiling of $T_1(e,f)$. So a statement about all triangles and all tiles is exchanged for the absence of tilings of one explicit two-parameter family, Theorem 1.2. The exchange is legitimate because the list of types is complete and because a prime number of tiles forces the scale to be one.

Theorem 1.2. No count can prove it. The same tile tiles $T_M(e,f)$ for all $M \geq f$, and a quantity that depends on the scale periodically or polynomially takes at scale one the value it takes at those scales (Proposition 7.7). The proof uses the order of the tiles: which tile lies next to which. Each side of the target is a word in the letters a, b, c, and the length of the side leaves few words. At a point of a side the angles of the tiles add up to π, which with α/π irrational leaves two figures, and this orients the tiles of the word one after the other. The step that carries the argument into the interior is a partition lemma (Lemma 2.3): a segment that is a union of edges on one side and is closed off at both ends is a union of whole edges on the other side as well. Since b is not a sum of other edges, and kb for $k<f$ is a sum of edges in one way only, a tile laid on such a segment forces the tile across it. One follows the forced tiles from a corner until a point is reached where no figure is possible. For the members $(1,f)$, $f \geq 3$, the walk runs along the base; for the others it starts at the apex.

Theorem 1.3. For a given N the classification and the arithmetic leave finitely many pairs (Δ, T) with explicit sides. Most of them are tiled by a known construction or fail a further arithmetic condition. Each of the others is decided by an exhaustive search, and the statement "no tiling exists" is exchanged for "this finite tree is a complete case distinction and every leaf is dead". The tree branches at a convex corner of the region not yet covered, where a tile of any tiling must have a corner, with an edge along the boundary: six placements. That these six cases are exhaustive is a theorem of the library, and each node of the tree, written in exact arithmetic over a real quadratic field, is checked by the Lean kernel. The search program that found the tree is not trusted and its floating-point verdict is not used.

Theorem 1.5. The tilings are explicit layouts of scaled tiles and of parallelograms made of two tiles, written in one skew coordinate system in which all vertices are integer points. The verification of a layout is exchanged for three finite checks: each piece lies in the target, any two pieces are separated by a line, and the numbers of tiles add up to the number that the area requires. The third check replaces the proof that the pieces cover the target, since pieces with disjoint interiors inside the target whose areas add up to its area leave no open set uncovered.

Theorem 1.6 and Corollary 1.7. Let $k<f$ equal tiles lie side by side with their sides c on s. Above them the tile lattice is forced row by row: each row of the lattice is a segment closed off at both ends, the partition lemma applies, and jb has one representation for $j<f$. The induction ends at a chord of the lattice triangle that cannot be covered, so a run of fewer than f such tiles cannot be bounded on both sides in the ways that the angles at the ends of s allow. A count of the possible neighbours of a run then shows that some run has length at least f. For a run of exactly f tiles the induction stops at its last chord, because $fb=(f^2-1)a$ has a second

representation; this is what the tilings of Theorem 1.5 show. The corollary follows by counting edges on a side modulo f (Section 7.3).

Theorem 1.8 and Corollaries 1.9 and 1.10. The proof for $e = 1$ uses two properties of $e = 1$, and the proof for all members replaces both. The first is an inequality between sides. The lattice triangle above a run of $k < f$ tiles is bounded on the right by the line through the right end of the run along the side a of the last tile, and the induction needs that every lattice point on this line is passed by a tile lying beyond the line. For $e = 1$ this is immediate: the tile that closes the run on the right lays an edge b or c on that line, the lattice points on it are at distances $a, 2a, \dots, ka$ from the run, and $ka < b$, so this one edge passes all of them. For $e \geq 2$ the inequality fails: $3a > b$ for the tile $(10, 21, 25)$, and $a > b$ for $(6, 5, 9)$. The general proof replaces the one long edge by the partition lemma applied to that line. The segment from the right end of the run to a lattice point is a union of $j < f$ edges a of lattice tiles; it is clamped at its lower end; and ja is a sum of edges only as $a + \cdots + a$ when $j < f$. So if no tile beyond the line passed the lattice point, the edge b or c of the closing tile would be one of j edges a. The cost is that the clamping at the lower end has to be carried through the induction as part of its hypothesis; at each step it is supplied by the side s or by an edge of a lattice tile already placed. The second property concerns angles: where the formal proof for $e = 1$ uses the inequality $3\alpha < \beta$, which is false for $(6, 5, 9)$, the general proof uses only the list of the ways in which angles of tiles fill a given angle, which is the same for every member (Lemma 2.7). Corollary 1.9 then follows by the count modulo f used for $e = 1$. Corollary 1.10 concerns the side of $W_M(e, f)$ that starts at the corner 2α. There the tile beside the first tile of a run can be its mirror image, a case that does not arise at a corner β or $\alpha + \beta$. A second induction over the same lattice excludes it for runs of at most e tiles. The bound e comes from the relation $ec = fa$: a row of e lattice edges c can also be covered by f edges a, and the induction does not control that covering (Section 7.4).

1.4. **Standing of the statements.** Every numbered statement carries one of three marks.

- F (formal). The statement is a theorem of the accompanying Lean 4 library, proved from Mathlib without `sorry` and with no axiom beyond `propext`, `Classical.choice` and `Quot.sound`. Appendix A gives the declaration. The mark says that the Lean kernel has checked a proof of the formal statement. It does not say that an independent reader has compared the formal statement with the mathematical one: this comparison has been made within this work, and not yet by an independent reader.
- W (written). The statement has a written proof in this paper and is not formalized.
- S (search). The statement is the outcome of a computer search or count that has not been turned into a certificate checked by the Lean kernel.

No statement marked F or W depends on a statement marked S. For a statement marked F the paper gives the architecture of the proof, in at most five steps with pointers, and writes out an argument in full only when it is short. Results of other authors are attributed in the statement or directly after it, also when the formal proof is ours.

1.5. **Organisation.** Section 2 contains the definitions and the classification of the tiles. Section 3 is on the signed-direction functional and proves Theorem 1.4. Section 4 is on Theorems 1.1 and 1.2, Section 5 on equilateral targets, Section 6 on Theorem 1.3, and Section 7 on the base-β targets above scale one. Section 8 states what is open, and Appendix A describes the formalization and contains the map from the numbered statements to the Lean declarations.

2. Definitions and the classification

2.1. **Dissections.**

Definition 2.1. A *triangle* is the convex hull of three affinely independent points of the plane. A *dissection* of a triangle Δ into N triangles is a family $t_1, \dots, t_N$ of triangles with $t_1 \cup \cdots \cup t_N = \Delta$ whose interiors are pairwise disjoint. It is a dissection *into N copies of T*, or an *N-tiling of Δ by T*, if every t_i is the image of T under an isometry of the plane. A positive integer N is a *tile count* if such a dissection exists for some Δ and some T.

Isometries include reflections, and nothing is assumed on how the tiles meet. These are the definitions of the Lean library (Section A.1). The sides of the tile are a, b, c and the opposite angles α, β, γ. An edge of a tile of length a is an *a-edge*, and a tile *shows* the angle α at a point if the point is its vertex of angle α; likewise for the other letters. An edge of length a joins the vertices of angles β and γ.

Proposition 2.2 (Dissection basics; F)**.** *Let $t_1, \dots, t_N$ be a dissection of Δ.*

(i) $\operatorname{Area}(\Delta) = \sum_i \operatorname{Area}(t_i)$*; for N copies of T,* $\operatorname{Area}(\Delta) = N \operatorname{Area}(T)$.

(ii) *At a point v of Δ the angles of the tiles containing v add up to 2π if v is interior to Δ, to π if v is in the relative interior of a side, and to the angle of Δ if v is a corner. A tile contributes its angle at v if v is one of its vertices and π if v lies in the relative interior of one of its edges.*

(iii) *Each side of Δ is the union of the edges of tiles that lie on it, and these edges follow one another along the side without overlap.*

By (iii) a side of the target of length L, tiled by a tile with sides a, b, c, gives nonnegative integers with $n_a a + n_b b + n_c c = L$; the sequence of the letters along the side is its *word*. The next lemma is the interior version of (iii).

Lemma 2.3 (Partition lemma; F)**.** *In a dissection of Δ, let $[P, Q]$ be a segment whose open part lies in the interior of Δ and contains no interior point of a tile, and let σ be one of the two sides of its line. Suppose that no tile lying on the side σ has an edge on the line that contains P or Q in its relative interior. Then the edges on the line of the tiles on the side σ that meet the open segment are contained in $[P, Q]$, and their lengths add up to $|PQ|$.*

The hypothesis at an end P holds, in particular, when the line just beyond P leaves Δ or enters the interior of a tile; we then say that the segment is *clamped* at P.

2.2. Classical families.

Proposition 2.4 (Classical families; F)**.** *If a triangle Δ is cut into N copies of T, then for every $k \geq 1$ the triangle $k\Delta$ is cut into $k^2 N$ copies of T. The numbers k^2, $3k^2$, $6k^2$ and $x^2 + y^2$ with $k, x, y \geq 1$ are tile counts.*

Proof. The lines parallel to the sides cut $k\Delta$ into k^2 triangles congruent to Δ. A triangle is a dissection of itself; an equilateral triangle is cut into three congruent triangles from its centre and into six by its medians; and the altitude of a right triangle with legs x, y cuts it into two triangles similar to it with ratios $x/\sqrt{x^2+y^2}$ and $y/\sqrt{x^2+y^2}$, which are cut into x^2 and y^2 congruent triangles. □

2.3. **The types of tiles.** The pairs (Δ, T) for which a tiling can exist were classified by Laczkovich [11, 12]. We use the classification in the organisation of Beeson [5]. If all angles of T are rational multiples of π, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [5, Thm. 3 and Table 3]. If some angle of T is not a rational multiple of π, and Δ is similar to T or T has a right angle, then N is a square or a sum of two squares [1, 22]. For a tile with an angle outside $\mathbb{Q}\pi$ the form in which the classification is used here is a theorem of the library.

Proposition 2.5 (Types of tiles; Laczkovich, in Beeson's organisation; F)**.** *Let Δ be cut into N copies of T, where some angle of T is not a rational multiple of π and Δ is not similar to T. Then for some $\alpha \notin \mathbb{Q}\pi$ one of the following holds.*

(R) *T has angles $(\alpha, \frac{\pi}{2}-\alpha, \frac{\pi}{2})$, and two angles of Δ are equal to α, or two are equal to $\frac{\pi}{2}-\alpha$.*
(D) *T has angles $(\alpha, \pi-3\alpha, 2\alpha)$, and two angles of Δ are equal to α.*
(S) *T has angles $(\alpha, \frac{2\pi}{3}-\alpha, \frac{\pi}{3})$, and Δ is equilateral.*
(H) *T has angles $(\alpha, \beta, \frac{2\pi}{3})$ with $\beta = \frac{\pi}{3}-\alpha$, and Δ is equilateral, or has the angles $(\alpha, \alpha, \alpha+3\beta)$ or $(\beta, \beta, \beta+3\alpha)$, or the angles of one of*

$$F_1 = (\alpha, \alpha+\beta, \alpha+2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha+\beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha+\beta),$$

or of the triangles F_1', F_3', F_4' obtained from these by exchanging α and β.
(T) *T has angles $(\alpha, \beta, 2\alpha+\beta)$ with $3\alpha+2\beta = \pi$, and Δ has the angles $(\beta, \beta, 3\alpha)$, $(\alpha+\beta, \alpha+\beta, \alpha)$, $(2\alpha, \beta, \alpha+\beta)$, $(2\alpha, \alpha, 2\beta)$ or $(\alpha, \alpha, \alpha+2\beta)$.*

The proof counts the angles of the tiles at the vertices of the tiling and at the corners of Δ; with an irrational angle the resulting equations between integers leave finitely many cases. The shapes of Δ in (H) that are not equilateral are those of Zhang [25].

In case (H) we always write (a, b, c) for the sides opposite $(\alpha, \beta, \frac{2\pi}{3})$, so that $c^2 = a^2+ab+b^2$. A *primitive triple* is a triple of positive integers with this relation and $\gcd(a, b) = 1$; then a, b, c are pairwise coprime and $3 \nmid c$.

2.4. **The tiles with $3\alpha+2\beta = \pi$.** A triangle with angles (α, β, γ), $\gamma = 2\alpha+\beta$ and $3\alpha+2\beta = \pi$, has $\gamma = \frac{\pi+\alpha}{2}$ and $\beta = \frac{\pi-3\alpha}{2}$, so by the law of sines

$$\frac{a}{c} = 2\sin\frac{\alpha}{2} =: s, \qquad \frac{b}{c} = 1-s^2.$$

If $s = e/f$ in lowest terms, the tile is similar to $(ef,\ f^2-e^2,\ f^2)$ with $1 \le e < f$. Conversely, for such e, f the angle α of this triangle has $\sin\frac{\alpha}{2} = \frac{e}{2f} \in (0, \frac{1}{2})$ rational, and α/π is irrational by Niven's theorem.

Definition 2.6. Let $1 \le e < f$ be coprime integers, $(a, b, c) = (ef,\ f^2-e^2,\ f^2)$, and $M \ge 1$ an integer. The triangle $T_M(e, f)$ has sides Mf^3, Mf^3, $Me(3f^2-e^2)$ and angles $\beta, \beta, 3\alpha$. The triangle $W_M(e, f)$ has sides $Mf(f^2-e^2)$, $Me(2f^2-e^2)$, Mf^3 and angles β, 2α, $\alpha+\beta$ opposite these sides. A *tiling* of $T_M(e, f)$ or of $W_M(e, f)$ is a tiling by copies of (a, b, c); it has $M^2(3f^2-e^2)$, respectively $M^2(2f^2-e^2)$, tiles. For $M \ge 1$ we write $\tau(e, f, M)$ for the statement: *$T_M(e, f)$ has a tiling if and only if $M \ge f$.*

The side $Mf(f^2-e^2) = fMb$ of $W_M(e, f)$ is the side b of a copy of the tile at scale fM. Attached along it, this copy completes $W_M(e, f)$ to $T_M(e, f)$, and the numbers of tiles agree: $M^2(2f^2-e^2) + f^2M^2 = M^2(3f^2-e^2)$. So a tiling of $W_M(e, f)$ gives a tiling of $T_M(e, f)$. A tiling of $T_M(e, f)$ need not arise in this way, because the common side need not be a union of edges; so the absence of a tiling of $W_M(e, f)$ does not exclude a tiling of $T_M(e, f)$.

Lemma 2.7 (Vertex figures; F)**.** *Let T have angles (α, β, γ) with $3\alpha+2\beta = \pi$, $\gamma = 2\alpha+\beta$, $\alpha \notin \mathbb{Q}\pi$. If tiles showing p angles α, q angles β and r angles γ fill an angle π at a point, then $(p, q, r) = (3, 2, 0)$ or $(1, 1, 1)$; in particular at most one of them shows γ. At a corner of angle β there is one tile, showing β; at a corner of angle 3α there are three tiles, each showing α.*

Proof. With $\beta = \frac{\pi-3\alpha}{2}$ and $\gamma = \frac{\pi+\alpha}{2}$, $p\alpha+q\beta+r\gamma = \frac{1}{2}\big((2p-3q+r)\alpha + (q+r)\pi\big)$. Setting this equal to π, β or 3α and using that α/π is irrational gives two linear equations in each case, whose nonnegative solutions are the ones listed. □

The same computation at a corner of angle $\alpha+\beta$ gives two tiles, showing α and β.

3. The signed-direction functional

In this section T has angles $(\alpha, \beta, \gamma) = (\alpha, \frac{\pi}{3} - \alpha, \frac{2\pi}{3})$ with $\alpha \notin \mathbb{Q}\pi$ and sides (a, b, c), and Δ is cut into N copies of T.

3.1. **Definition and cancellation.** Fix a reference direction along an edge of one tile. The angles of T lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$, and two tiles that meet at a point have edge directions there which differ by sums of angles of tiles and of π; since Δ is connected, every edge of every tile has a direction $j\frac{\pi}{3} + k\alpha$ with integers j, k. As $\frac{\pi}{3}$ and α are linearly independent over $\mathbb{Q}$, the direction determines k, and j modulo 6. Put

$$f_\alpha(\theta) = (-1)^j \quad \text{for } \theta = j\tfrac{\pi}{3} + k\alpha, \qquad f_\beta(\theta) = (-1)^{j'} \quad \text{for } \theta = j'\tfrac{\pi}{3} + k'\beta.$$

Both are defined on all directions that occur, $f_\beta = (-1)^k f_\alpha$, and $f(\theta + \pi) = -f(\theta)$ for $f \in \{f_\alpha, f_\beta\}$, since $\pi = 3 \cdot \frac{\pi}{3}$. The *weight* of a directed segment of length L and direction θ is $Lf(\theta)$, and $\Phi_f(\partial P)$ denotes the sum of the weights of the three sides of a triangle P, directed counterclockwise.

Lemma 3.1 (Cancellation; F)**.** *For $f \in \{f_\alpha, f_\beta\}$: $\sum_{i=1}^{N} \Phi_f(\partial t_i) = \Phi_f(\partial\Delta)$.*

Proof. Cut every edge of every tile at every point where another edge meets it. The pieces obtained do not overlap partially, and since the weight is linear in the length, the sums do not change. A piece in the interior of Δ is covered once from each of its two sides, by tiles whose boundaries run along it in opposite directions, so its two weights cancel. A piece on the boundary of Δ is covered by one tile, in the counterclockwise direction of $\partial\Delta$. □

Lemma 3.2 (Value on a tile; F)**.** *For every tile t of the tiling, $\Phi_{f_\alpha}(\partial t) = \pm(c + a - b)$ and $\Phi_{f_\beta}(\partial t) = \pm(c + b - a)$.*

Proof. Let V_1, V_2, V_3 be the vertices of a positively oriented tile, counterclockwise, with angles α, β, γ. The directed edges V_1V_2, V_2V_3, V_3V_1 have lengths c, a, b, and the direction turns by $\pi - \beta$ and then by $\pi - \gamma$. In the coordinates $j\frac{\pi}{3} + k\alpha$ the angle π adds 3 to j, β adds 1 and γ adds 2; so j takes the values $j_0, j_0 + 2, j_0 + 3$ and the sum is $(-1)^{j_0}(c + a - b)$. In the coordinates $j'\frac{\pi}{3} + k'\beta$ the angle β adds 0 and γ adds 2; so j' takes the values $j_0', j_0' + 3, j_0' + 4$ and the sum is $(-1)^{j_0'}(c - a + b)$. A reflected tile is traversed in the opposite sense, which changes the sign. □

Corollary 3.3 (Integrality and parity; F)**.** *The numbers*

$$M_\alpha = \frac{\Phi_{f_\alpha}(\partial\Delta)}{c + a - b}, \qquad M_\beta = \frac{\Phi_{f_\beta}(\partial\Delta)}{c + b - a}$$

are integers, and $M_\alpha \equiv M_\beta \equiv N \pmod 2$.

Proof. By Lemmas 3.1 and 3.2 each of them is a sum of N signs. □

A change of the reference direction changes M_α and M_β at most by a sign. The two conditions are separate necessary conditions for a tiling.

3.2. **The isosceles obstruction.**

Lemma 3.4 (Non-integrality; F)**.** *Let a, k be coprime positive integers, $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then k does not divide $a + b - c$.*

Proof. We have $a < c < a + b$. If $k \mid a + b - c$, then $c \equiv a \pmod k$ and $c = a + kt$ with $1 \le t \le k - 1$. Substituting and dividing by k gives $a(2t - k) = k(k - t)(k + t)$. So $k \mid a(2t - k)$, hence $k \mid 2t$, and $2 \le 2t \le 2k - 2$ gives $2t = k$. Then the left side is 0 and the right side is positive. □

Proposition 3.5 (Rationality from the two integers; F)**.** *Let $a, b, c > 0$ with $c^2 = a^2 + ab + b^2$ and let $P \neq 0$ be real. If $P^2 b(a + 2b)$, $P(c - a - b)$ and $P(c + a + b)$ are integers, then a/b and c/b are rational.*

Proof. Write N, M_1, M_2 for the three integers. Then $M_1 M_2 = P^2(c^2 - (a + b)^2) = -P^2 ab$, so $N + M_1 M_2 = 2P^2 b^2 \neq 0$ and $a/b = -2M_1 M_2/(N + M_1 M_2)$. Also $M_2 - M_1 = 2P(a + b) \neq 0$ and $M_1 + M_2 = 2Pc$, so $c/(a + b) = (M_1 + M_2)/(M_2 - M_1)$. □

Proof of Theorem 1.4. Let Δ be isosceles, not equilateral, and tiled by a prime number N of copies of T. By Proposition 2.5 its base angles are α or β: the other shapes of (H), and a triangle similar to T, are scalene or equilateral, because distinct integer combinations of α and $\frac{\pi}{3}$ are distinct angles. Exchanging the names of α and β if necessary, let the base angles be α. Since $2\cos\alpha = (a + 2b)/c$, the sides of Δ are λc, λc, $\lambda(a + 2b)$ for a real $\lambda > 0$, and by Proposition 2.2(i)

$$N = \frac{\lambda^2(a + 2b)}{b}.$$

Take the reference direction along the base. The legs have the directions $\pi - \alpha$ and $\pi + \alpha$, with $j = 3$ in the coordinates of f_α and $j' = 2$, $j' = 4$ in those of f_β. Hence $\Phi_{f_\alpha}(\partial\Delta) = \lambda(a + 2b - 2c)$ and $\Phi_{f_\beta}(\partial\Delta) = \lambda(a + 2b + 2c)$, and the identities $(c - a - b)(c + a - b) = b(a + 2b - 2c)$ and $(c + a + b)(c + b - a) = b(a + 2b + 2c)$ give

$$M_\alpha = \frac{\lambda(c - a - b)}{b}, \qquad M_\beta = \frac{\lambda(c + a + b)}{b}.$$

By Corollary 3.3 these are integers, and Proposition 3.5 with $P = \lambda/b$ shows that the sides of T are commensurable. Scale the figure so that (a, b, c) is a primitive triple. The sides λc and $\lambda(a + 2b)$ of Δ are sums of edges, hence integers, so λ is rational. A common prime factor of c and $a + 2b$ would divide $(a + 2b)^2 - c^2 = 3b(a + b)$; it does not divide b or $a + b$, since then it would divide a and b, and it is not 3, since $3 \nmid c$. So c and $a + 2b$ are coprime and λ is an integer. Now $Nb = \lambda^2(a + 2b)$ with $\gcd(b, a + 2b) = 1$ gives $b \mid \lambda^2$ and $N = (\lambda^2/b)(a + 2b)$. As N is prime and $a + 2b \geq 3$, we get $\lambda^2 = b$. Then $M_\alpha = (c - a - b)/\lambda$ is an integer, with $b = \lambda^2$ and $\gcd(a, \lambda) = 1$, against Lemma 3.4. □

Both functionals are used: together they give the rationality of the tile, and on the target with base angles β their roles are exchanged. The formal statement of Theorem 1.4 has no hypothesis of rationality on the tile.

3.3. **The numbers of tiles on the other targets.**

Proposition 3.6 (Forms of N; F)**.** *Let (a, b, c) be a primitive triple and let Δ be cut into N copies of a triangle similar to (a, b, c), where a, b, c are opposite $\alpha, \beta, \frac{2\pi}{3}$.*

(i) *If Δ has the angles of F_2, F_3 or F_4, then $N = N_0 k^2$ for an integer $k \geq 1$, with $N_0 = (a + 2b)(2a + b)$, $3(a + b)(a + 2b)$, $(a + b)(2a + b)$ respectively.*

(ii) *If Δ has the angles of F_1 and $b = de^2$ with d squarefree, then $N = dw^2(a + b)$ for an integer $w \geq 1$.*

(iii) *If Δ is similar to the tile, N is a square.*

(iv) *If Δ is equilateral with side S, in the unit in which the tile is (a, b, c), then S is an integer, $N = S^2/(ab)$, and $c + a - b$ and $c + b - a$ divide $3S$.*

Proposition 3.7 (The isosceles and F_1 targets; arithmetic; F)**.** *Let (a, b, c) be a primitive triple and $N \geq 1$. There are integers $k \geq 1$ and $M_\alpha \equiv M_\beta \equiv N \pmod 2$ with*

$$Nb = k^2(a + 2b), \qquad M_\alpha(c + a - b) = k(a + 2b - 2c), \qquad M_\beta(c + b - a) = k(a + 2b + 2c)$$

if and only if $N = m^2b(a+2b)$ *for an integer* $m \geq 1$. *The corresponding statement holds with the equations of the isosceles target with base angles* β *and* $N = m^2a(2a+b)$, *and with those of* F_1 *and* $N = m^2b(a+b)$.

The three equations are the area and the two integers of Corollary 3.3 on the isosceles target with base angles α at scale k. So the numbers of tiles that these conditions allow on the isosceles targets and on F_1 are exactly the numbers of the families constructed by Zhang [25], who realizes them for m in certain residue classes from a bound on. The conditions are necessary and not sufficient: $33 = 3 \cdot 11$ has this form for the tile $(5, 3, 7)$ and is not a tile count (Theorem 1.3). Which small members are realized is part of what is open (Section 8).

4. The prime case

4.1. The primes that arithmetic does not exclude.

Proposition 4.1 (F). *Let* $p > 3$ *be a prime. There are coprime integers* $1 \leq e < f$ *with* $p = 3f^2 - e^2$ *if and only if* $p \equiv 11 \pmod{12}$.

Proof. Let $p = 3f^2 - e^2$. If $3 \mid e$ then $3 \mid p$; so $p \equiv -e^2 \equiv 2 \pmod 3$. The integers e, f are not both even, and not both odd since then $p \equiv 2 \pmod 4$; in the two remaining cases $p \equiv 3 \pmod 4$. Conversely let $p \equiv 11 \pmod{12}$. By quadratic reciprocity 3 is a square modulo p, so p is represented by a binary quadratic form of discriminant 12. There are two classes of such forms, represented by $x^2 - 3y^2$ and $3y^2 - x^2$, and the first represents only primes $\equiv 1 \pmod 3$. So $p = 3y^2 - x^2$ with positive integers x, y, coprime because p is prime. If $x \geq y$ then $y < x < \sqrt{3}\, y$, and $(|2x - 3y|,\ 2y - x)$ is another representation with a smaller second entry; so the representation with y least has $x < y$. □

A prime has at most one such representation (`ThinHole.rep_unique`).

4.2. The case analysis.

Let N be a prime with $N \equiv 3 \pmod 4$ and $N > 3$, and suppose that Δ is cut into N copies of T. The formal proof of Theorem 1.1 excludes this in five steps.

1. *All angles of* T *are rational multiples of* π. Write the angles as $a_k\pi/n$ with $\gcd(a_0, a_1, a_2) = 1$. A valuation argument on the area identity gives $N \mid n$. For a prime $p \geq 7$ dividing n the space of rational linear relations attached to $(n; a_0, a_1, a_2)$ is the trivial one, and this forces Δ to be similar to T or of one exceptional shape. The first is step 2, and the second is excluded for primes $N \geq 5$ (`LemmaSAll.r0c`).
2. Δ *is similar to* T. Then a prime N is 3 or $\not\equiv 3 \pmod 4$ (`Row2.theoremS_perm`).
3. *Otherwise* T and Δ are of one of the types of Proposition 2.5.
4. *Types* (R), (D), (S), (H) *and two targets of* (T) are excluded by the area and the boundary weights: Table 2.
5. *The three remaining targets of* (T). The boundary weights show that a/c is rational, so the tile is $(ef, f^2 - e^2, f^2)$ up to scale, and a prime N forces the scale of the target to be one. Each case gives a tiling of $T_1(e, f)$ for some member, which Theorem 1.2 excludes.

For the last two rows recall that $T_1(e, f)$ is $W_1(e, f)$ with a tile at scale f attached. It is also the union of two tiles at scale f, placed at its base corners, and of the triangle between them, which has legs fb, base eb and angles $(\alpha + \beta, \alpha + \beta, \alpha)$; a tiling of a triangle of the last shape by a prime number of tiles is carried to this triangle by a homothety of integer ratio followed by the subdivision of each tile. For this shape there is also a direct argument, which does not use Theorem 1.2 (Proposition 6.4(iii)). Step 1 uses cyclotomic fields and the non-vanishing of $L(1, \chi)$ for Dirichlet characters, from Mathlib.

tile and target	reason	Lean declaration
(R)	N is even	`Row3.RP.row3_RP_even_any_label`
(D)	N is not prime	`Row4.not_prime_any_label`
(S), equilateral	N is not a prime > 3	`Row5.theoremE_sixty_full`
(H), equilateral	N is not a prime > 3	`Row5.theoremE_twoPiThree_full`
(H), isosceles	Theorem 1.4	`ThmIsoFull.thmIso`
(H), F_1, F_1'	the product $M_\alpha M_\beta = N$ and the sides	`F1FreeProof.f1_free_statement`
(H), F_2 to F_4'	$N = N_0 k^2$, N_0 composite	`R8F2to4Bridge.r8_reduction_statement`
(T), $(\alpha, \alpha, \alpha + 2\beta)$	N is not prime	`Row11.prime_excluded_any_label`
(T), $(2\alpha, \alpha, 2\beta)$	$N = k^2(2f^2 - e^2)(3f^2 - e^2)$	`rMinus4`
(T), $(\beta, \beta, 3\alpha)$	Δ is $T_1(e, f)$	`ScaleBaseBetaFull.scaleBaseBeta`
(T), $(2\alpha, \beta, \alpha + \beta)$	Δ is $W_1(e, f)$; add a tile at scale f	`RWStatementProof.rwStatement`
(T), $(\alpha + \beta, \alpha + \beta, \alpha)$	add two tiles at scale f	`RAStatementProof.raStatement`

TABLE 2. Steps 4 and 5 of the case analysis for a prime $N \equiv 3 \pmod 4$, $N > 3$. The files are in `lean/Erdos634/Draft/`.

Remark 4.2 (The literature on the last three rows)**.** In [3] the base-β targets are treated in Theorem 14, which asserts that $\gcd(a, c)$ divides the colouring number of the tiling; for a tiling of $T_m(e, f)$ this is the assertion $f \mid m$. The 99-tiling of $T_3(1, 2)$, the triangle $(24, 24, 33)$, by the tile $(2, 3, 4)$ is a counterexample; it is kernel-checked (Proposition 6.1). The preprint [2] refers to [3] for the tiles with $3\alpha + 2\beta = \pi$. The exclusion of $T_1(e, f)$ and $W_1(e, f)$ for every member, and with it Theorem 1.1, was first published by Paliy [18], with a different proof.

4.3. **The base-β targets at scale one.** Throughout, $(a, b, c) = (ef, f^2 - e^2, f^2)$ with coprime $1 \le e < f$. The proof of Theorem 1.2 has this architecture.

1. *Normal form.* A dissection whose target and tile have the stated sides is carried by an isometry to a fixed position; the vertex figures are those of Lemma 2.7; the sides are words in a, b, c (`BaseBetaWalks.lean`, `E1NormalForm.normal_form`).
2. *The members* $(1, f)$, $f \ge 3$. The base carries exactly one b-edge, with f a-edges and one c-edge, or two c-edges, or $2f$ a-edges. Each word is excluded by the figures at its junctions and the partition lemma applied to segments above the base (`E1BaseWord.e1_family_f_ge_3_of_sides`).
3. *The pierced corner.* Lemma 4.3 below, which holds for every member.
4. *The members* $(f - 1, f)$, $f \ge 2$. Here $a + c = fb$, and only here. This places the corner V of Lemma 4.3 one length a above the base. A tile showing α or β at V against the continuation of the ray would have a vertex below the base. In the remaining arrangement, γ against the ray and α against the edge of t_2, a segment of length $a - b$ is left, and $a - b$ is not a sum of edges (`R1Final.r1_no_tiling`).
5. *The members with* $2 \le e \le f - 2$. The pierced corner forces a patch of translates of the apex tiles along the ray of Lemma 4.3. At the boundary of the patch the remaining angle allows two continuations, and every continuation that leaves the lattice is excluded by an induction along a row, by a reflection lemma, or by a march along two converging lines of the lattice; an induction along the other inner ray ends the argument (`TheoremHCongruent.theorem_H_of_sides`).

The three families of steps 2, 4 and 5 cover all members; $(1, 2)$ falls under step 4. The assembled statement is `BaseBetaFullUnc.basebeta_full_unc`. For steps 4 and 5 the proof is the formal one: the statements of the intermediate lemmas are written out, but there is no continuous written proof. This is part of the standing recorded after Theorem 1.1.

Lemma 4.3 (The pierced corner; F)**.** *Let a triangle with angles* $(\beta, \beta, 3\alpha)$ *be tiled by copies of* (a, b, c)*, and suppose that the edges of the two legs at the apex are* c*-edges. Let* t_2 *be the middle*

one of the three tiles at the apex. Then one of the two edges of t_2 at the apex is its c-edge and lies against the b-edge of a neighbouring apex tile, and the vertex V of t_2 at the far end of this c-edge lies in the relative interior of an edge of another tile. The tiles that have a vertex at V on the side of t_2, other than t_2, show the angles $\{\alpha,\gamma\}$ or $\{\alpha,\alpha,\alpha,\beta\}$.

At scale one the hypothesis holds, since a leg of $T_1(e,f)$ carries no b-edge: every side of the target carries a c-edge (`Dissection.congruentDissection_gammatrap`), and a sum of edges of length f^3 that contains a c contains no b (`BaseBetaWalks.side_no_b`).

Proof. By Lemma 2.7 three tiles show α at the apex, and the edges of a tile at its vertex of angle α are b and c. The outer two lay c on the legs, so they turn their b-edges inward, and t_2 turns one b-edge and one c-edge inward. On the ray where the c-edge of t_2 meets a b-edge, the b-edge ends at distance b from the apex, in the relative interior of the c-edge. Beyond that point the c-edge of t_2 has the length $c-b=e^2$ left, and the tiles on its other side lay consecutive edges along it, starting at that point. But e^2 is not a sum of edges: $ef>e^2$ and $f^2>e^2$, and $n(f^2-e^2)=e^2$ would give $f^2\mid n+1$, hence $e^2\geq f^2-1$ and $f^2-e^2=1$, which is impossible. So one of these edges passes through the end V of the c-edge. The tiles at V on the side of t_2 fill the angle π, of which t_2 takes β, and by Lemma 2.7 the others show $\{\alpha,\gamma\}$ or $\{\alpha,\alpha,\alpha,\beta\}$. □

4.4. **The formal statement and its standing.** In the notation of Section A.1, the Lean theorem behind Theorem 1.1 is

$$\begin{aligned}\texttt{theoremP_6}:\quad &\forall N\in\mathbb{N},\ \forall D:\texttt{CongruentDissection}\ N,\\ &N\text{ prime}\ \to\ N \bmod 4=3\ \to\ 3<N\ \to\ \texttt{False},\end{aligned}$$

and Theorem 1.1 itself is `PrimeCharacterization.prime_tile_count_iff`, with Proposition 2.4 and the theorem on sums of two squares for the other direction. Neither has a hypothesis beyond those displayed. Both are in the aggregate `Erdos634.All`, whose import closure contains no search certificate. One module of the proof of Proposition 2.5, the enumeration of the shapes of the target, is generated by a program and checked by Lean.

Within this work the definitions have been compared with Definition 2.1, the two formal statements with Theorems 1.1 and 1.2, and the formal proofs of the cases of Section 4.2 and of the three families of Section 4.3 have been read a second time, separately from their writing. No independent reader has yet done so, and for $e\geq 2$ there is no continuous written proof of Theorem 1.2. Until such a check has been made, Theorem 1.1 is a formally verified statement whose match with the mathematical statement has not been confirmed independently, and we make no further claim.

5. Equilateral targets: commensurable sides

Let an equilateral triangle of side S be cut into N copies of a tile of type (S) or (H), with sides (a,b,c) opposite $(\alpha,\beta,\frac{\pi}{3})$, respectively $(\alpha,\beta,\frac{2\pi}{3})$, and $\alpha\notin\mathbb{Q}\pi$.

Proposition 5.1 (The pair of a tiling; F)**.** *For a tile of type* (S) *there are integers $m_1>m_2\geq 1$, with $m_1\leq N$ and $m_1\equiv m_2\equiv N \pmod 2$, such that*

$$m_1(a+b-c)=3S=m_2(a+b+c),\qquad S^2=Nab,\qquad m_1m_2=3N.$$

For a tile of type (H) *there are integers $1\leq s,t\leq N$ with $s\equiv t\equiv N \pmod 2$ such that*

$$s(c+a-b)=3S=t(c+b-a),\qquad S^2=Nab,\qquad st=3N.$$

For type (H) the integers are M_α and M_β of Corollary 3.3: the three sides of the target differ in direction by $\frac{2\pi}{3}$, so they have the same weight. Type (S) is analogous. From the displayed equations,

$$(m_1+m_2)^2-16N=\Big(\frac{2N(a-b)}{S}\Big)^2,\qquad (t-s)^2+16N=\Big(\frac{2N(a+b)}{S}\Big)^2. \tag{1}$$

Theorem 5.2 (Laczkovich; Beeson and Zhang; F)**.** *In the situation above, for a tile of either type, there are integers $j\ge 1$ and $p,q\ge 0$ with $ja=pb+qc$, and integers $j'\ge 1$, $p',q'\ge 0$ with $j'b=p'a+q'c$. Consequently the sides a,b,c are pairwise commensurable.*

For type (S) the commensurability is a theorem of Laczkovich [12, Thm. 3.3]. For type (H) the relations and the commensurability are Lemma 3.5 and Theorem 5.2 of Beeson and Zhang [7]. The proof formalized is their degree count, for type (S) with the roles of two vertex figures exchanged, and with a different last step for type (H). What is added is the formal proof.

Architecture, for type (S). A *vertex* is a vertex of some tile, and its figure (x,y,z) counts the tiles showing α,β,γ there.

1. Since $\alpha\notin\mathbb{Q}\pi$, every figure has $x=y$; it is $(0,0,1)$ at a corner of the target, $(1,1,1)$ or $(0,0,3)$ at a vertex where the angles add up to π, and $(3,3,0)$, $(2,2,2)$, $(1,1,4)$ or $(0,0,6)$ at a vertex where they add up to 2π.
2. Every tile has one angle of each kind, so $\sum_V(z_V-x_V)=0$. With $\mathcal{O}$ the set of vertices of figure $(3,3,0)$ or $(1,1,4)$ and $\mathcal{T}$ the set of interior vertices of figure $(0,0,3)$, this gives $|\mathcal{O}|\ge|\mathcal{T}|+1$.
3. Suppose no relation $ja=pb+qc$ holds. Call a directed a-edge from a vertex *mismatched* if another tile has an edge of length b or c from the same vertex along the same ray. On its side of the line the a-edges continue until a first point H, its *head*, where the next edge is not an a-edge; by the hypothesis no vertex of the other side lies on the way, so H is an interior vertex where the angles add up to π.
4. Different mismatched edges have different heads; a mismatched edge starts at every vertex of $\mathcal{O}$, where the number of a-edges is odd, and at every head of figure $(1,1,1)$.
5. Counting mismatched edges by their heads and by their starting points gives $|\mathcal{O}|\le|\mathcal{T}|$, against step 2.

The second relation follows by exchanging α and β, and the two relations with Proposition 5.1 give the commensurability. □

Corollary 5.3 (Pairs with a non-square discriminant; F)**.** *In Proposition 5.1 the integer $(m_1+m_2)^2-16N$, respectively $(t-s)^2+16N$, is a perfect square. So for every N, a pair (m_1,m_2) or (s,t) for which this integer is not a square is the pair of no tiling.*

Proof. By Theorem 5.2 the sides a,b,c are commensurable, and S is commensurable with them by the first equation of Proposition 5.1. So the right-hand sides of (1) are squares of rational numbers, and the left-hand sides are integers. □

For a given N there are finitely many pairs, and each pair determines the tile up to scale, with S. The corollary removes the pairs whose tile would have incommensurable sides without any search. For type (S) the formal proof of Theorem 5.2 does not use Theorem 1.2. The hypotheses of the theorem are satisfied by kernel-checked tilings with 1440 tiles, for type (S), and 1215 tiles, for type (H), in the library.

6. The tile counts up to 200

6.1. **The tilings.**

Proposition 6.1 (Tilings; F)**.** *The following dissections exist.*

(i) $T_M(1,2)$, *with sides* $(8M, 8M, 11M)$, *and* $W_M(1,2)$, *with sides* $(6M, 7M, 8M)$, *into* $11M^2$ *and* $7M^2$ *copies of* $(2,3,4)$, *for every* $M \geq 2$.
(ii) *The triangle* $(16, 28, 33)$ *into* 77 *copies and the triangle* $(24, 24, 42)$ *into* 84 *copies of* $(2,3,4)$.
(iii) *The triangle* $(36, 36, 63)$ *into* 189 *copies of* $(2,3,4)$.
(iv) *The triangle* $(21, 55, 56)$ *into* 88 *copies and the triangle* $(39, 49, 55)$ *into* 143 *copies of* $(3,5,7)$.
(v) *The triangle* $(45, 81, 84)$ *into* 126 *copies of* $(6,5,9)$.

Moreover $7m^2$ *and* $11m^2$ *are tile counts if and only if* $m \geq 2$.

Each tiling of (ii) to (v), and the tilings of (i) for $M = 2, 3$, is a list of triangles with exact coordinates in a ring $\frac{1}{d}\mathbb{Z}[\sqrt{K}]$, for which the Lean kernel checks that every triangle has the sides of the tile, that it lies in the target, that any two of the triangles are separated by the line of an edge of one of them, and that the areas add up to the area of the target (Section A.2). Attributions: tilings with $28 = 7 \cdot 2^2$ and with 77 tiles were constructed by Beeson [3]; tilings of the triangles of (iii) and of $(21, 55, 56)$ were exhibited by Harries [10]; the triangle of (v) is $W_3(2,3)$, the case $(2,3)$ of Theorem 7.1. The 143-tiling is shown in Figure 1. Its target has the angles of F_2 for the tile $(3,5,7)$, and $143 = (a+2b)(2a+b)$ is the least number that Proposition 3.6 allows there; the numbers of tiles on this shape for this tile are exactly $143k^2$, $k \geq 1$ (`F2Row357.f2_row_357`). Zhang [25] constructs $143m^2$ for m in the semigroup generated by 5 and 7, and the ledger [20] announces every $m \geq 2$. We know of no earlier tiling with 143 tiles: Harries [10] lists 143 as unresolved with this pair, and the ledger has no entry for the number. This is a statement about what we have read, not a claim of priority. The last sentence of the proposition uses Theorem 1.1 for $m = 1$.

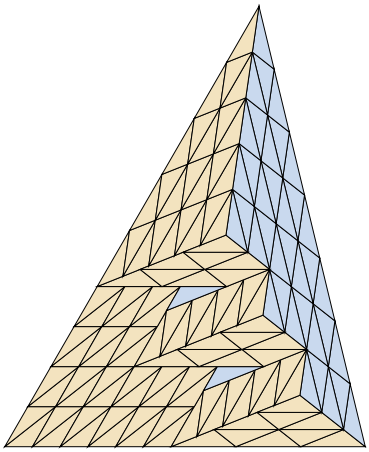


FIGURE 1. The triangle $(39, 49, 55)$ cut into 143 copies of $(3,5,7)$ (Proposition 6.1(iv)). The two shades are the two orientations of the tile.

With Proposition 2.4 this gives the 105 values of Table 1: the classical families, the numbers $7m^2$ and $11m^2$ with $m \geq 2$, and 77, 84, 88, 126, 143, 189.

6.2. **The exclusions.** The 95 values $N \leq 200$ outside Table 1 are the 23 primes $\equiv 3 \pmod 4$ between 7 and 199, excluded by Theorem 1.1, and 72 composite numbers. For a composite N the formal proof runs through the classification: the tiles with angles in $\mathbb{Q}\pi$, the targets similar to the tile and the right-angled tiles are closed by arithmetic statements of the library for the given

N; for each other type of Proposition 2.5 the equations of Sections 3 and 5 and their analogues for type (T) leave a finite list of pairs (Δ, T) with explicit sides, computed by the kernel; and each pair of the list is excluded by a general theorem or by a search certificate (Section A.2). The general theorems used are Theorem 1.2, Corollary 5.3 and Propositions 6.4 and 6.5. The values fall into the following groups.

- The 24 composite values up to 87: 14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 46, 51, 55, 56, 57, 60, 62, 66, 69, 70, 76, 78, 86, 87. Certificates are used at 15, 56, 60, 70 and 76.
- 32 values for which the arithmetic closes every case: 91, 93, 94, 95, 102, 110, 111, 114, 115, 118, 119, 123, 129, 133, 134, 138, 141, 142, 152, 155, 158, 159, 165, 166, 174, 177, 182, 183, 186, 187, 190, 195.
- 168 and 198, where the arithmetic leaves only pairs of type (S) on the equilateral triangle with a non-square discriminant (Corollary 5.3).
- 92, 124, 140, 171, 172, 188, with two to four pairs (Δ, T) each, of the types (T) and (D), every one excluded by a certificate or by Proposition 6.4.
- 105: two tiles of type (S) on the equilateral triangle of side 105 and two pairs on the shape F_1, the tiles $(8, 7, 13)$ on $(56, 91, 105)$ and $(16, 5, 19)$ on $(80, 95, 105)$; four certificates. That 105 is not a tile count was announced earlier by Paliy [15], with certificates for the same four pairs; the Lean theorem confirms it.
- 120: the tiles with angles in $\mathbb{Q}\pi$ (Proposition 6.5), the tile $(8, 13, 15)$ of type (S) on the equilateral triangle of side 120, and the tile $(7, 8, 13)$ on the triangle $(56, 104, 120)$ of shape F_1; two certificates, of 65 710 and 88 484 nodes.
- 132, 154, 156, 161, 184, each with one to three pairs, every pair excluded by a certificate. The largest certificate for each value is: the tile $(5, 3, 7)$ on $(42, 42, 66)$ for 132 (233 117 nodes); $(8, 7, 13)$ on $(91, 91, 154)$ for 154 (63 895); $(12, 7, 16)$ on $T_2(3, 4) = (128, 128, 234)$ for 156 (234 239); $(12, 7, 16)$ on $(112, 112, 161)$ for 161 (345 746); $(7, 8, 13)$ on $(104, 104, 184)$ for 184 (165 736), whose second pair is $(10, 21, 25)$ on $W_2(2, 5) = (184, 210, 250)$ (71 173). The exclusion of 154 was announced earlier by Paliy [20]. For 132, 156, 161 and 184 we know of no earlier claim: Harries [10] lists them as unresolved or dependent on an incomplete induction, and the ledger has no entry.

Formal content of Theorem 1.3. One Lean theorem states the equivalence of Theorem 1.3 for all $N \leq 200$ outside 132, 135, 143, 154, 156, 161, 184, and in particular for all $N \leq 131$ (`Spectrum200g.tileCount_iff_le_200g`, `tileCount_iff_le_131`). Six further theorems state that 143 is a tile count and that 132, 135, 154, 156, 161, 184 are not (`N143.tileCount_143`, `N132F.not_tileCount_132`, `N135F.not_tileCount_135`, `Spectrum200Cond4.decided_154_156`, `N161F.not_tileCount_161`, `N184F.not_tileCount_184`). They are not combined into one formal statement, because a module that imports several certificates together exceeds the memory available; the theorem that combines them is proved with the six exclusions as hypotheses (`Spectrum200Excl.tileCount_iff_le_200_of_excl`), and each of the six hypotheses is a theorem, checked in its own aggregate. □

6.3. **The value** 135**.**

Proposition 6.2 (F)**.** 135 *is a tile count if and only if some equilateral triangle has a tiling by* 135 *copies of a tile of type* (H) *whose pair* (s, t) *of Proposition 5.1 is* $(15, 27)$ *or* $(27, 15)$*.*

A tiling with this pair has its tile similar to $(3, 5, 7)$ and the side 45 in the unit of the tile; this identification is elementary and is not part of the formal statement. So the question is whether the equilateral triangle of side 45 can be cut into 135 copies of $(3, 5, 7)$. Every other case for 135 is closed in Lean: the tiles with angles in $\mathbb{Q}\pi$ by Proposition 6.5, the pair $(9, 45)$ by Corollary 5.3, and two pairs of the types (D) and (T) by certificates.

Theorem 6.3 (F). *The equilateral triangle of side* 45 *cannot be cut into* 135 *copies of the triangle* $(3, 5, 7)$. *Consequently* 135 *is not a tile count.*

Proof by certificate. A search places the tiles at the lowest uncovered point, one corner at a time, and records why every branch fails. Its tree has 1 326 256 nodes in exact arithmetic over $\mathbb{Q}(\sqrt{3})$, with 81 629 learned sets of tiles that no tiling contains. Only the half of the tree whose first tile shows the angle α on the base is written out; the other half is refuted through the mirror image of the target. The tree is cut into 2766 files, and the Lean kernel checks each node against the geometry of the library. The second sentence follows with Proposition 6.2. □

Paliy [20] has announced that $15m^2$ is a number of tiles for this tile exactly when $m \geq 4$, with a computer-assisted exclusion of $m = 3$. Theorem 6.3 reproduces that exclusion with a certificate of our own that the Lean kernel has checked; the claim is his. For $m = 4$ the tiling of the equilateral triangle of side 60 by 240 copies, which is also his, is kernel-checked here (`N240.equilateral_60_by_357`); for $m = 1, 2$ the numbers 15 and 60 are not tile counts (Theorem 1.3).

6.4. Statements for infinitely many N.

Proposition 6.4 (Isosceles targets with base angles $\alpha + \beta$; F). *Let* $0 < e < f$ *be coprime,* $(a, b, c) = (ef, f^2 - e^2, f^2)$, *and* $t \geq 1$ *an integer.*

(i) *If the triangle with sides* tfb, tfb, teb *is cut into copies of* (a, b, c), *then* $teb = Pa+Qb+Rc$ *with integers* $P, Q \geq 0$ *and* $R \geq 2$.

(ii) *For every* $k \geq 1$ *the triangle with sides* $2(k + 1)(2k + 1)$, $2(k + 1)(2k + 1)$, $2k(2k + 1)$ *cannot be cut into copies of* $(k(k + 1), 2k + 1, (k + 1)^2)$.

(iii) *Let* $3\alpha + 2\beta = \pi$ *with* $\alpha \notin \mathbb{Q}\pi$. *If an isosceles triangle with base angles* $\alpha + \beta$ *is cut into* N *congruent triangles with angles* $(\alpha, \beta, 2\alpha + \beta)$, *then* $N = t^2(f^2 - e^2)$ *for integers* $t \geq 2$ *and coprime* $0 < e < f \leq te$. *In particular* N *is neither prime nor squarefree.*

Part (i) is the conclusion of [3, Lem. 6] for this side. As usually quoted, that every side of the target carries two c-edges, the lemma fails for the dissection of $(4, 6, 8)$ into four copies of $(2, 3, 4)$; what is added is a proof under explicit hypotheses. Part (ii) is the case $(e, f, t) = (k, k + 1, 2)$, where the base has no representation with $R \geq 2$, and is used for $N = 124$, 140, 172 and 188. Part (iii) contains the prime case of this target, and its proof does not use Theorem 1.2.

Proposition 6.5 (Angles in $\mathbb{Q}\pi$ at $15s^2$ and $30s^2$; F). *Let* $N = ds^2$ *with* $d \in \{15, 30\}$ *and* $s \geq 1$. *No triangle can be cut into* N *congruent triangles all of whose angles are rational multiples of* π.

For these N the statement is contained in the result of [5] quoted in Section 2.3; the proposition is a formal proof of this case. It uses an identity between sines, which is divided by s^2 and refuted, for the ten tiles that a rank criterion does not exclude, by certificates in cyclotomic fields for 9 280 pairs of a tile and a target.

Proposition 6.6 (Twice a prime; F). *Let* p *be a prime. Then* $2p$ *is a tile count if and only if* $p = 3$ *or* $p \not\equiv 3 \pmod 4$.

For $p \equiv 3 \pmod 8$ the exclusion follows from Theorem A of Paliy [17]. The formal proof uses Theorem 1.2.

Theorem 6.7 (Squarefree N prime to 3, by residue modulo 8; F). *Let* N *be squarefree with* $3 \nmid N$, *and suppose that* N *is a tile count.*

(i) $N \not\equiv 3 \pmod 8$.

(ii) *If* $N \equiv 7 \pmod 8$, *then* $N = ab$ *or* $N = (a+2b)(2a+b)$ *for positive integers* a, b *with* a^2+ab+b^2 *a square. Below* 455 *the only such tile count is* 143.
(iii) *If* $N \equiv 5 \pmod 8$, *then* N *is a sum of two squares, or* $N = (2q^2-p^2)(3q^2-p^2)$ *for coprime* $0 < p < q$, *or* $N = (f^2-e^2)(2f^2-e^2)$ *for coprime* $1 \le e < f$. *Below* 517 *the only such tile count that is not a sum of two squares is* 77.

Most of (i) reproduces a result of Paliy: the reduction of every tiling with such an N to a tiling of some $T_1(e,f)$ with $N = 3f^2-e^2$ is the general clause of Theorem A of [16], which excludes every N of the class for which this equation has no solution. Added here are a formal proof of the reduction and the case in which the equation has a solution, by Theorem 1.2. All three parts use Theorem 1.2 and inherit its standing.

Remark 6.8 (Squarefree N in general)**.** There is a decidable arithmetic condition on a squarefree N, with one clause for each case of the classification, which implies that N is not a tile count; the implication is a Lean theorem (`AllN3.not_tile_of_conditions3`).

Computation 6.9 (S)**.** Of the 60 789 squarefree N with $7 \le N \le 10^5$, 13 600 are sums of two squares, 46 898 satisfy the condition of Remark 6.8, and 291 are neither. Those of the 291 that are below 1000 are 7, 11, 15, 21, 30, 33, 70, 77, 105, 143, 154, 161, 231, 322, 329, 330, 345, 354, 385, 455, 465, 506, 517, 546, 777, 806, 897, 935, 966.

The count is by a program that transcribes the condition and is not checked by the kernel value by value. Of the values listed, those up to 161 are decided by Theorems 1.1 and 1.3.

7. The base-β targets above scale one

7.1. Existence.

Theorem 7.1 (The seed at scale f; Beeson, Paliy; F)**.** *Let* $1 \le e < f$ *be integers. The triangle with sides* $f^2(f^2-e^2)$, $ef(2f^2-e^2)$, f^4 *can be cut into* $f^2(2f^2-e^2)$ *copies, and the triangle with sides* f^4, f^4, $ef(3f^2-e^2)$ *into* $f^2(3f^2-e^2)$ *copies, of* $(ef,\ f^2-e^2,\ f^2)$. *For coprime* e, f *these triangles are* $W_f(e,f)$ *and* $T_f(e,f)$.

Proof. Write $a = ef$, $b = f^2-e^2$, $c = f^2$, $P = 3f^2-e^2$. Let g_1, g_2 be unit vectors with $g_1 \cdot g_2 = -\cos\beta = -eP/(2f^3)$, and use the coordinates $(x,y) \mapsto xg_1 + yg_2$. The parallelogram $[0,a]\times[0,c]$ has a diagonal of length b, since $a^2+c^2-2ac\cos\beta = b^2$; it is two copies of the tile. Put

$$O=(0,0),\quad D=(ab,0),\quad E=(0,-a^2),\quad B=(ab,-a^2),$$
$$A=(ab,bc),\quad C=(-ac,-a^2),\quad Z=(-2ac,-a^2).$$

A computation with the quadratic form x^2+y^2-xyeP/f^3 gives $|AZ| = |AB| = c^2$ and $|ZB| = aP$, so AZB is the second triangle of the statement. The triangles OCA and ODA are similar to the tile with ratio b, the triangle OEC with ratio a, and the triangle CZA with ratio c; each is cut into tiles by the lines parallel to its sides. The parallelogram $ODBE$ is $[0,ab]\times[-a^2,0]$, which is $b\times e^2$ cells $[0,a]\times[0,c]$. These five pieces lie in AZB and are separated by the lines OA, OC, CA, $x=0$ and $y=0$. Their numbers of tiles add up to

$$2b^2+a^2+2be^2+c^2 = f^2(3f^2-e^2),$$

the number that the area of AZB requires, so they cover it. The first four pieces form the triangle ACB, with sides $|AC| = bc$, $|CB| = a(b+c)$, $|BA| = c^2$ and $2b^2+a^2+2be^2 = f^2(2f^2-e^2)$ tiles. □

The four pieces of ACB are the dissection that Paliy's notes [20] credit to Beeson [3, Thm. 11], and the fifth piece is the "beta lift" of the same notes. The proof above, with all seven points at integer coordinates in one frame, is the form in which it is formalized.

Architecture of the proof of Theorem 1.5. Put $Q = 2f^2 - e^2$.

1. *Seed.* $W_f(e, f)$ has a tiling (Theorem 7.1).
2. *Collar.* For $T > f - e$, a tiling of W_T extends to a tiling of W_{T+e}: the triangle W_{T+e} is the union of a translate of W_T, two rectangles of a skew frame tiled by parallelograms $a \times b$, four triangles similar to the tile, and two parallelograms tiled along other lattices of the tile. The nine pieces are separated by lines and contain $Q(T + e)^2$ tiles in all (`CollarG.collar_nat`).
3. *Further seeds.* For $1 \le t < e$ the triangle W_{f+t} is the union of two truncated scaled tiles, one scaled tile, and a hexagon cut into $e+t$ strips of triangles and parallelograms and one more scaled tile; the numbers of tiles add up to $Q(f+t)^2$ (`Residue.wTilable_residue`).
4. *All scales.* Every $M \ge f$ is $f + t + ie$ with $0 \le t < e$ and $i \ge 0$, and $f + t > f - e$; so steps 1 to 3 give W_M (`Residue.wTilable_ge`).
5. *Lift.* $T_M(e, f)$ is $W_M(e, f)$ with a tile at scale fM attached (Definition 2.6; `Residue.betaTilable_ge`).

In the formal proof the parameters of the layouts are real numbers, every separating inequality is certified by a polynomial with nonnegative coefficients in the parameters, and integrality is used only for the numbers of rows and columns. □

The collar of step 2 and the seeds of step 3 are Paliy's [20]. For $e = 1$ step 3 is empty.

7.2. Scale two for $f \le 4$.

Proposition 7.2 (F). *For $(e, f) \in \{(1, 3), (2, 3), (1, 4), (3, 4)\}$ the triangle $T_2(e, f)$ has no tiling by any number of copies of $(ef, f^2 - e^2, f^2)$. The triangle $T_2(1, 2) = (16, 16, 22)$ is cut into 44 copies of $(2, 3, 4)$.*

The four exclusions are search certificates, of 56 016, 31 709, 56 629 and 234 239 nodes, for the targets $(54, 54, 52)$, $(54, 54, 92)$, $(128, 128, 94)$ and $(128, 128, 234)$.

7.3. The members $(1, f)$ below the scale f.

Architecture of the proof of Theorem 1.6. Place s on a horizontal line with Δ above it. A tile with an edge on s has one of six types, named by the edge and the angles at its left and right ends: $A_1 = (a; \beta, \gamma)$, $A_2 = (a; \gamma, \beta)$, $B_1 = (b; \alpha, \gamma)$, $B_2 = (b; \gamma, \alpha)$, $C_1 = (c; \alpha, \beta)$, $C_2 = (c; \beta, \alpha)$.

1. *Tools.* The figures of Lemma 2.7, the partition lemma, and three facts of arithmetic: jb is a sum of edges only as $b + \cdots + b$ for $1 \le j < f$; a representation of jc with $j < f$ contains no b; and $ja < b$ for $j \le f - 1$ (`SideRun.rep_jb`, `SideRun.rep_qc`).
2. *Lattice.* Above consecutive tiles $X_0, \ldots, X_{k-1}$ of type C_1 consider the lattice with the vectors of the sides c and b of X_0; its triangles are copies of the tile, and X_i is one of them.
3. *Run lemma.* For $k < f$ the following cannot hold together: the tile beside X_0 across its b-edge shows β at the left end of the run, and the right end of the run is a corner of angle β of Δ, or the next tile on s shows γ or β there. The proof is an induction over the rows of the lattice triangle above the run: each row is a segment clamped at both ends, the partition lemma and the arithmetic of step 1 force its tiles, and the last chord of the triangle cannot be covered (`SideRun.Ctx.run_dies`).

4. *Rules.* If all runs of C_1 and of C_2 on s are shorter than f, then by step 3 and its mirror image a maximal run that follows a tile of type A_1 or B_1, or starts at the left corner, is followed by a tile of type A_1 or B_1, directly or after a second run (`SideRun.Setup.side_rules`).
5. *The word.* The first tile of s is not A_2 or B_2, the last one is A_2 or C_1, and A_1, B_1 are never followed by A_2, B_2. With step 4, after every tile of type A_1 or B_1 there is a later one, which is impossible on a finite side (`SideRun.Setup.side_theorem'`).

Consecutive tiles of one type C_1 or C_2 are translates of one another, and f of them cover a length $fc = f^3$ of s. □

The hypotheses of Theorem 1.6 are satisfied by the tiling of $W_f(1, f)$ of Theorem 7.1 (`SideRun.side_theorem_witness`).

Proof of Corollary 1.7. Let $(a, b, c) = (f, f^2 - 1, f^2)$. If a side of integer length n carries x, y, z edges of lengths a, b, c, then $n = xf + y(f^2 - 1) + zf^2 \equiv -y \pmod f$, so

$$y \equiv -n \pmod f. \tag{2}$$

(a) Let $M < f$ and suppose $W_M(1, f)$ has a tiling. The side of length $n = M(2f^2 - 1)$ lies opposite the angle 2α, so its corners have the angles $\alpha+\beta$ and β. By (2), $y \equiv M \pmod f$, hence $y \geq M$, because $0 < M < f$. Then $zf^2 \leq M(2f^2 - 1) - M(f^2 - 1) = Mf^2$ and $z \leq M < f$. But by Theorem 1.6 this side carries at least f tiles with their side c on it. For $M \geq f$ a tiling exists by Theorem 1.5.

(b) Let $2M \leq f$ and suppose $T_M(1, f)$ has a tiling. Both corners of the base have the angle β, and its length is $n = M(3f^2 - 1)$. As before $y \geq M$, so $zf^2 \leq M(3f^2 - 1) - M(f^2 - 1) = 2Mf^2$ and $z \leq 2M$, while Theorem 1.6 gives $z \geq f$. So $2M = f$ and $z = f$. Then $xf + y(f^2 - 1) = M(3f^2 - 1) - f^3 = M(f^2 - 1)$ with $y \geq M$ gives $y = M$ and $x = 0$: the base carries no a-edge, and its f tiles with c on the base are the tiles of Theorem 1.6, all of one type. But the tile at each base corner shows β there and its edges at the corner are a and c (Lemma 2.7); without a-edges the first tile of the base is of type C_2 and the last one of type C_1. This is a contradiction. □

At $M = 1$ part (b) gives Theorem 1.2 for the members $(1, f)$, $f \geq 2$, by a second route. For $T_2(1, f)$ part (b) covers $f \geq 4$, and $f = 3$ is Proposition 7.2; so $\tau(1, f, 2)$ holds for every $f \geq 2$ (`E1ScaleTwo.thresholdAt_1_f_2`). For $f/2 < M < f$ the count allows a run of f tiles on the base and the argument gives nothing. For $e \geq 2$ the formal proof does not apply: the first two arithmetic facts of step 1 hold for every member, but $(f - 1)a < b$ fails when $e \geq 2$. Section 7.4 describes a proof that does not use this fact.

7.4. **Every member below the scale f.** Theorem 1.8, Corollaries 1.9 and 1.10 and Proposition 7.3 are theorems of the Lean library (Table 3); below we give the architecture of the proofs and the short arguments in full. The hypotheses of the formal version of Theorem 1.8 are shown to hold for the tilings of Theorem 1.5, for every member. For $e = 1$, Theorem 1.8 is Theorem 1.6 and Corollary 1.9 is contained in Corollary 1.7, which are formal.

Changes to the architecture for Theorem 1.8. The five steps are those of the proof of Theorem 1.6 in Section 7.3, with the tile $(a, b, c) = (ef, f^2 - e^2, f^2)$ and the following changes.

1. *Tools.* If $n = xa + yb + zc$ with integers $x, y, z \geq 0$, then $ye^2 \equiv -n \pmod f$, and e is invertible modulo f. From this: jb is a sum of edges only as $b + \cdots + b$ for $1 \leq j < f$; a representation of jc with $j < f$ contains no b; and ja is a sum of edges only as $a + \cdots + a$ for $1 \leq j < f$. The third fact replaces $ja < b$. At $j = f$ the first and the third fail, since $fb = (f - e)(a + c)$ and $fa = ec$.

3. *Run lemma.* The hypothesis at the right end R of the run is: R is a corner of angle β of Δ, or the tile beside the last tile of the run across its side a shows α at R, so that it lays an edge b or c on the line of that side. This line bounds the lattice triangle on the right. Its segment from R to a lattice point consists of fewer than f edges a of lattice tiles and is clamped at R; by the partition lemma and the third fact of step 1, a tile beyond the line has the lattice point in the relative interior of an edge. For $e = 1$ this was obtained from $ja < b$. In the induction the same hypothesis is needed with R replaced by a lattice point Z of an inner line, and it then includes that the line is clamped below Z; the clamp is the side s or an edge of a lattice tile that ends at Z. Wherever a wedge is filled, only the figures of Lemma 2.7 and the corresponding lists for the angles α, $\alpha+\beta$, 2α, 3α and γ are used; no inequality between α and β and none between a and b is used.

Steps 2, 4 and 5 are unchanged. □

Proof of Corollary 1.9. Let $(a, b, c) = (ef, f^2-e^2, f^2)$. If a side of integer length n carries x, y, z edges of lengths a, b, c, then $n \equiv -ye^2 \pmod f$. For $n = Me(2f^2-e^2)$ and for $n = Me(3f^2-e^2)$ we have $n \equiv -Me^3 \pmod f$; since e is invertible modulo f, this gives

$$y \equiv Me \pmod f. \tag{3}$$

(a) Let $Me < f$ and suppose that $W_M(e, f)$ has a tiling. The side of length $n = Me(2f^2-e^2)$ lies opposite the angle 2α, so its corners have the angles $\alpha+\beta$ and β. By (3), $y \geq Me$, because $0 < Me < f$. Then $zf^2 \leq Me(2f^2-e^2) - Me(f^2-e^2) = Mef^2$ and $z \leq Me < f$. But by Theorem 1.8 this side carries at least f tiles with their side c on it.

(b) Let $2Me < f$ and suppose that $T_M(e, f)$ has a tiling. Both corners of the base have the angle β, and its length is $n = Me(3f^2-e^2)$. As before $y \geq Me$, so $zf^2 \leq Me(3f^2-e^2) - Me(f^2-e^2) = 2Mef^2$ and $z \leq 2Me < f$, against Theorem 1.8. □

For the members with $f \leq 40$ a count of the words, made by a program and not checked by the kernel, shows that the two conditions are the limit of this argument: for $M < f$ some word on the side of part (a) has at least f letters c when $Me \geq f$, and some word on the base when $2Me \geq f$.

Corollary 1.10 uses the third side of $W_M(e, f)$, of length Mfb, which runs from the corner 2α to the corner $\alpha+\beta$. Theorem 1.8 does not apply to it. The statement that takes its place has a second alternative.

Proposition 7.3 (Sides from a corner 2α or 3α; F)**.** *Let $1 \leq e < f$ be coprime integers and $(a, b, c) = (ef,\ f^2-e^2,\ f^2)$. Let s be a side of a triangle Δ that is cut into copies of (a, b, c), from a corner A' of angle 2α or 3α to a corner Q' of angle β or $\alpha+\beta$; place s horizontally with A' on the left and Δ above, and use the types of Section 7.3. Then one of the following holds.*

(i) *s carries f consecutive tiles of type C_1 or f consecutive tiles of type C_2.*

(ii) *The tiles on s begin at A' with exactly k tiles of type C_1, where $e+1 \leq k \leq f-1$; the first of them and the tile beside it at A' are mirror images in their common edge of length b; and the k tiles are followed by the corner Q' or by a tile of type A_2, B_2 or C_2.*

If $a < b$ and A' has the angle 2α, then $k \geq e+2$ in (ii)*. For the members $(f-1, f)$ alternative* (ii) *is empty.*

The proof first shows, by the rules of step 4, that s carries no tile of type A_1 or B_1 if all runs are shorter than f; so s begins with a run of $k < f$ tiles of type C_1 whose right end satisfies the hypothesis of the run lemma. The tile beside the first tile at A' shows α there. If it lays its side c along the first tile, the induction of the run lemma applies and gives a contradiction. Otherwise it is the mirror image of the first tile, and a second induction over the lattice triangle shows that this is impossible for $k \leq e$. That induction uses that a row of fewer than e lattice

edges c is covered from above only by edges c; a row of e edges c can be covered by f edges a, because $ec = fa$. For $k = e+1$, $a < b$ and a corner 2α, the covering by edges a is excluded because one of its tiles would leave Δ across the second side at A'. Whether alternative (ii) occurs at all is not known to us.

Proof of Corollary 1.10. Let $(a, b, c) = (ef, f^2 - e^2, f^2)$, and let the side of $W_M(e, f)$ of length $n = Mfb$ carry x, y, z edges of lengths a, b, c. It lies opposite the angle β, so it runs from the corner 2α to the corner $\alpha + \beta$. Since $n \equiv 0 \pmod f$, we have $y = mf$ with an integer $m \geq 0$, and dividing $n = xef + mfb + zf^2$ by f gives $xe + zf = (M - m)b$. Put $M' = M - m$. If $M' = 0$, then $z = 0$. Let $M' \geq 1$. Modulo f the equation reads $xe \equiv -M'e^2$, so $x \equiv -M'e \equiv M'(f - e)$.

(a) Let $M(f - e) \leq e$. Then $0 < M'(f - e) \leq e < f$, so $x \geq M'(f - e)$, and $zf \leq M'(f^2 - e^2) - M'(f - e)e = M'(f - e)f$. Hence $z \leq M'(f - e) \leq e$: the side carries at most e tiles with their side c on it. So it carries neither f consecutive tiles of type C_1 or C_2 nor $e + 1$ tiles of type C_1, against Proposition 7.3.

(b) For $M \leq f - 1$ this is part (a) with $e = f - 1$, where $M(f - e) = M \leq e$. For $M \geq f$ a tiling exists by Theorem 1.5.

(c) Here $(a, b, c) = (15, 16, 25)$ and $n = 160$. The solutions of $15x + 16y + 25z = 160$ with $5 \mid y$ are $(x, y, z) = (9, 0, 1)$, $(4, 0, 4)$, $(2, 5, 2)$ and $(0, 10, 0)$, so $z \leq 4$. Since $a < b$ and the side starts at the corner 2α, Proposition 7.3 requires 5 consecutive tiles with their side c on it, or an initial run of $k \geq e + 2 = 5$ tiles of type C_1. □

Parts of these corollaries were known. At $M = 1$ they give no new statement: $T_1(e, f)$ and $W_1(e, f)$ have no tiling by Theorem 1.2. The pairs $W_2(2, 3)$, $W_2(3, 4)$, $W_2(5, 6)$ and $W_2(2, 5)$ were excluded before by search certificates, in the formal exclusions of 56, 92, 188 and 184 (Section 6.2); for them the corollaries are a second argument. The scale-one statements were proved first by Paliy [18]. The ledger [20] states that every side of a tiled $W_M(e, f)$ or $T_M(e, f)$ carries two whole edges of length f^2, and for $W_2(f - 1, f)$ it records the forced word c, c, a, a on the side Mfb. We have not compared the proofs, and we make no claim of priority.

All arguments of this subsection concern one side of the target, and they exclude a target only when one of its sides is too short to carry f edges c. For $W_M(e, f)$ the two sides give $Me < f$ and $M(f - e) \leq e$; for $T_M(e, f)$ the legs are long enough at every scale and the base gives $2Me < f$. For the remaining pairs every side admits a word with f consecutive letters c, and an exclusion has to use two sides together or the interior of the target.

7.5. **The layer.**

Proof of Proposition 1.11. In this proof M denotes the scale of the triangle that contains the layer (the scale M' of the statement), so that part (a) reads: if $2M \geq f + 2$ and $T_{M-1}(1, f)$ has a tiling, so has $T_M(1, f)$. Let $(a, b, c) = (f, f^2 - 1, f^2)$. The height of $T_1(1, f)$ is $H_1 = f^3 \sin\beta$, which is f times the height $c \sin\beta$ of the tile over its side $a = f$. Hence a trapezoid of height H_1 with parallel sides u and v, or a triangle of height H_1 over a side u (with $v = 0$), has the area of $u + v$ tiles. All pieces below have the height H_1. From left to right along the base, the layer consists of:

- the tile at scale f, with its side fa on the base and its side fc on the left leg; it has f^2 tiles, and its third side, of length fb, gives the direction of the cuts between the following pieces;
- $x_a = (2M - f - 2)f + 1$ columns, each made of f parallelograms with a side a parallel to the base and a side b in the direction of the cuts; a column has width a and $2f$ tiles;
- M columns, each made of b parallelograms with a side b parallel to the base and a side a in the direction of the cuts; a column has width b and $2b$ tiles;

- the quadrilateral of six pieces of the collar of Theorem 1.5 for $e = 1$, which is a trapezoid with bottom $fc+(f-1)a$, top $(f-1)b$, left side fb in the direction of the cuts and right side fc on the right leg; it has $fc+(f-1)a+(f-1)b = 1+2fb$ tiles.

Consecutive pieces share a segment of length fb, since f sides b and b sides a both give $fb = ba$. The condition $x_a \geq 0$ is $2M \geq f+2$. The widths on the base add up to

$$fa + x_a a + Mb + fc + (f-1)a = \big(1 + (2M - f - 2) + M + f + 1\big)f^2 - M = M(3f^2 - 1),$$

the base of $T_M(1, f)$, and the widths on the upper line to $x_a a + Mb + (f-1)b = (M-1)(3f^2-1)$. The numbers of tiles add up to

$$f^2 + 2fx_a + 2bM + 1 + 2fb = 6Mf^2 - 3f^2 - 2M + 1 = (2M-1)(3f^2-1),$$

which is $M(3f^2-1) + (M-1)(3f^2-1)$, the number that the area of the layer requires. The part of $T_M(1, f)$ above the layer is a translate of $T_{M-1}(1, f)$.

(a) If $2M \geq f+2$, a tiling of $T_{M-1}(1, f)$ and the layer give a tiling of $T_M(1, f)$. If $2M \leq f+1$, then $2(M-1) \leq f$ and $T_{M-1}(1, f)$ has no tiling by Corollary 1.7(b). (b) By (a) the set of scales with a tiling is closed under $M \mapsto M+1$; it contains f by Theorem 1.5 and no M with $2M \leq f$. (c) follows from (b). □

The formal proof packs the same pieces: a translate of $T_{M'-1}$, the parallelograms, the six pieces of the trapezoid and the tile at scale f. The trapezoid of six pieces is Paliy's. The proposition also shows where an answer to Question 1.12 for $e = 1$ cannot come from: for $2M \geq f+2$ the part of $T_M(1, f)$ below the height H_1 has a tiling with a straight upper side, so no argument confined to that part excludes $T_M(1, f)$.

7.6. **Every member: a signed decomposition, a layer, and a strip of $W_M(e, f)$.** Both propositions are formal; part (b) of Proposition 7.4 and part (a) of Proposition 7.5 are stated in the form of the formal statements, which assert that the pieces lie in the trapezoid with disjoint interiors and, with a tiling of the smaller triangle, form a tiling of the larger one. The propositions are constructions and implications between tilings, and they decide no instance. A *wall* of a tiling is a segment that contains no interior point of a tile.

Proposition 7.4 (Every member: signed decomposition and layer; F)**.** *Let $1 \leq e < f$ be coprime integers, $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$, and $M \geq 1$. Let $T_M(e, f) = ABC$ with apex A and base BC, and let H_1 be the height of $T_1(e, f)$ over its base.*

(a) *Let Q', Q be the points of the base with $|BQ'| = |CQ| = Mef^2$, and let Z be the point of the ray AQ beyond Q with $|QZ| = Me^2f$. Then ABQ', ACQ and $AQ'Z$ are copies of the tile at scale fM; the triangle $Q'QZ$ is a copy of the tile at scale Me and lies on the other side of the base; and $T_M(e, f) \cup Q'QZ = ABQ' \cup AQ'Z \cup ACQ$, with disjoint interiors on each side. The triangle ABQ is a copy of $W_M(e, f)$, and $ABQ \cup Q'QZ = ABQ' \cup AQ'Z$.*

(b) *If $2f(M-e) \geq b$, the part of $T_M(e, f)$ above the line parallel to the base at the height eH_1 is a translate of $T_{M-e}(e, f)$, and the trapezoid below that line contains pieces with disjoint interiors, cut into $(2Me - e^2)(3f^2 - e^2)$ copies of the tile in all, which together with a tiling of that translate form a tiling of $T_M(e, f)$ in which the line is a wall.*

(c) *If $2fM \geq b$ and $T_M(e, f)$ has a tiling, so has $T_{M+e}(e, f)$. If $2fM \geq b$ and $k \geq 1$, then $T_M(e, f)$ has a tiling if and only if some tiling of $T_{M+ke}(e, f)$ has the chord parallel to the base at the height keH_1 as a wall.*

For the members $(f-1, f)$ the condition of (b) *is $M \geq f$.*

For $e = 1$ part (b) is the layer of Proposition 1.11. For the members $(f-1, f)$ the layer exists only at the scales $M \geq f$, so parts (b) and (c) relate no two scales below f. For the other members with $e \geq 2$ two scales below f are related only when $(f-e)^2 \geq 2f$.

The idea. For (a): two copies of the tile at scale fM fit into the base corners of $T_M(e,f)$, with their sides fMc on the legs, and between them an isosceles triangle with the angle α at A remains. A third copy at scale fM, laid against one of the two, covers this triangle and exceeds it beyond the base by a copy of the tile at scale Me; the lengths agree because $b+e^2=c$. So the target is three scaled tiles less one. A scaled tile is cut into tiles by the lines parallel to its sides. Hence a quantity that is additive over the tiles, or an invariant of the boundary with values in a group, takes on $T_M(e,f)$ the value it takes on this signed family of tiles, at every scale, and so does not exclude $T_M(e,f)$ at any scale; the same holds for $W_M(e,f)$. Unlike Proposition 7.7, this does not use the tilings at the scales $M \geq f$. An argument that uses an inequality satisfied by each tile is not covered by it. For (b): the layer is a row of pieces of one height, cut from one another by parallel lines. At one base corner lies the tile at scale ef. Then come $2Mf-(f^2+2ef-e^2)$ columns of ef parallelograms with a side a on the base, and eM columns of b parallelograms with a side b on the base; each parallelogram is two tiles, and the slant side of either kind of column has the length efb. At the other corner lies a trapezoid of six pieces: the four triangles and the two parallelograms of the collar in step 2 of the proof of Theorem 1.5. The condition $2f(M-e)\geq b$ says that the first number of columns is not negative. With these pieces the height cannot be less than eH_1: at a height gH_1 with $g<e$ a column with the side b on the base would consist of gb/e parallelograms, and b is prime to e. Part (c) follows by stacking layers, since the upper side of a layer is straight.

Proposition 7.5 (Every member: a strip of $W_M(e,f)$, and the scale $f-1$; F)**.** *Let $1\leq e<f$ be coprime integers and $(a,b,c)=(ef,\ f^2-e^2,\ f^2)$. For $M\geq 2$ let s be the side of length Mfb of $W_M(e,f)$, which joins the corners 2α and $\alpha+\beta$, and let s' be the chord parallel to s that cuts off a translate of $W_{M-1}(e,f)$ at the corner β.*

(a) *Let $M\geq 2$, and let z be an integer with $z\equiv Mf \pmod e$ and $f\leq z\leq Mb/f$. Then the trapezoid between s and s' is the union, with disjoint interiors, of lattice pieces each cut into copies of the tile, $(2M-1)(2f^2-e^2)$ in all; so a tiling of $W_{M-1}(e,f)$ gives one of $W_M(e,f)$ in which s' is a wall. Such an integer z exists if and only if $M(f-e)\geq f$.*
(b) *If $M(f-e)\geq f$ and $W_{M-1}(e,f)$ has a tiling, then $W_M(e,f)$ has a tiling.*
(c) *$W_M(e,f)$ has no tiling for every $M<f$ if and only if $W_{f-1}(e,f)$ has none.*
(d) *$W_{f-1}(e,f)$ has a tiling if and only if some tiling of $W_f(e,f)$ has the chord s' as a wall.*

Part (c) is a reduction in one direction. If $W_{f-1}(e,f)$ has no tiling, then no smaller scale has one; a tiling of $W_{f-1}(e,f)$ would decide nothing about the smaller scales. Part (c) uses Corollary 1.10(a), which is formal. For the members $(1,f)$ and $(f-1,f)$ the triangles $W_M(e,f)$ are settled (Corollaries 1.7 and 1.10), so parts (c) and (d) are new for $2\leq e\leq f-2$. The first targets $W_{f-1}(e,f)$ of these members are $W_4(2,5)$, with 736 tiles, and $W_4(3,5)$, with 656; both are open.

The idea. The strip consists of five pieces, each a union of triangles of a lattice tiling. On s, from the corner 2α, lie z tiles with their side c on s and then $x=(Mb-zf)/e$ tiles with their side a on s, so that $zc+xa=Mfb$. Above the first group are the f lowest rows of the lattice tiling of the tile at scale z; above the second group are x columns of e parallelograms with sides a and c. The two groups fit because $fa=ec$: in one direction, f edges a and e edges c have the same length. A tile at scale f, reflected in the bisector of the corner 2α, and a tile at scale e at the corner $\alpha+\beta$ close the two ends, and a parallelogram of $(M-1)f$ columns of f parallelograms with sides b and a fills the rest up to s'. The f rows require $z\geq f$. The count of edges on s in the proof of Corollary 1.10 gives $z\leq M(f-e)$ when $M(f-e)<f$, so that no such z exists then; and when $M(f-e)\geq f$ the interval from f to Mb/f contains e consecutive integers. For (c): if $W_M(e,f)$ has a tiling at a scale $M<f$, then $M(f-e)\geq e+1$

by Corollary 1.10(a), hence $(M+1)(f-e) \geq f+1$, and (b) applies at every larger scale up to $f-1$.

The trapezoid of six pieces in the layer is Paliy's, as for $e=1$, and the tilings at the scales $M \geq f$, to which Proposition 7.4(c) and Proposition 7.5(d) refer, are Paliy's and Beeson's (Theorem 1.5). Paliy's collar passes from $W_T(e,f)$ to $W_{T+e}(e,f)$ along the side of length $Me(2f^2-e^2)$. The strip of Proposition 7.5, along the side of length Mfb and in steps of one, was not found in the ledger [20] as read; this is an absence in one unrefereed source, and we make no claim of priority.

7.7. What is known on Question 1.12.

Proposition 7.6 (Reduction and complete members; F)**.** *Let $1 \leq e < f$ be coprime. Then $\tau(e,f,M)$ holds for $M=1$ and for every $M \geq f$, and it holds for every $M \geq 1$ if and only if $T_M(e,f)$ has no tiling for $2 \leq M < f$. Moreover:*

(i) *$\tau(1,2,M)$, $\tau(1,3,M)$ and $\tau(2,3,M)$ hold for every $M \geq 1$;*
(ii) *$\tau(1,4,M)$ and $\tau(3,4,M)$ hold for every $M \neq 3$;*
(iii) *$\tau(1,f,M)$ holds whenever $2M \leq f$ or $M=2$ or $M \geq f$.*

Proof. The first sentence is Theorems 1.2 and 1.5. The rest follows with Proposition 7.2 and Corollary 1.7. □

By Corollary 1.9(b), $\tau(e,f,M)$ also holds whenever $2Me < f$; this is formal as well.

Proposition 7.7 (Quantities determined backwards; F)**.** *Let $1 \leq e < f$ be coprime, let ι be a function on the positive integers with values in a set X, and let $z \in X$ be such that $\iota(M) = z$ whenever $T_M(e,f)$ has a tiling. Suppose that $\iota(M) = z$ follows from $\iota(M') = z$ for all $M' > M$. Then $\iota(M) = z$ for every M.*

Proof. By Theorem 1.5, $\iota(M) = z$ for $M \geq f$; induct downwards. □

The hypothesis holds for a periodic ι, for an ι with values in an abelian group and a vanishing iterated difference, and for an ι that satisfies a linear recurrence which can be solved for its first term; these three cases are Lean lemmas. Counts modulo an integer, additive functions of the edge vectors, areas, and the images of the boundary word of $T_M(e,f)$ in finite, nilpotent or linear quotients of the group generated by the edge steps modulo the boundaries of the tiles, in the sense of Conway and Lagarias [9], are of these three kinds; we do not write out this verification here, and it is not formalized. Such a quantity therefore takes the same value at $M=1$, where there is no tiling, as at the scales with a tiling, and cannot answer Question 1.12. An answer has to use the order of the tiles, as the proofs of Theorems 1.2 and 1.6 do.

8. What is open

We list what a determination of the set of tile counts still requires. We make no estimate of how far these items are from being settled.

8.1. An outside check of the prime case.

Theorems 1.1 and 1.2 are formal theorems whose statements have not been checked by an independent reader (Section 4.4). Such a reader has to compare the definitions of Section A.1 and the two formal statements with the mathematical ones; a continuous written proof of Theorem 1.2 for $e \geq 2$ does not exist. The exclusions of composite N that use Theorem 1.2 (Sections 6.2 and 6.4) inherit this.

8.2. **All N.** No description of the set of tile counts above 200 is proved or conjectured here. For each N the classification leaves finitely many pairs (Δ, T) with explicit sides, and the question is which of them have tilings. No family of pairs can be excluded as a whole, since each of the families below contains pairs with tilings. By type:

- *Angles in $\mathbb{Q}\pi$; Δ similar to T; a right angle.* Settled in the literature: these give the classical families [5, 1, 22].
- *Type* (D). N is not squarefree [4, Thm. 11.7]. For the values of Section 6 the remaining pairs of this type were excluded one at a time; no general statement is known to us.
- *Types* (S) *and* (H) *on the equilateral triangle.* By Corollary 5.3 only tiles with integer sides remain, finitely many for each N. Which of them tile is open; Laczkovich relates the possible tiles to rational points on elliptic curves [14]. For the tile $(3, 5, 7)$ the sides 15, 30 and 45 have no tiling and the side 60 has one.
- *Type* (H) *on the other targets.* The number of tiles is $m^2 N_0$ with N_0 given by Propositions 3.6 and 3.7. Zhang [25] constructs tilings for every sufficiently large m in certain residue classes, and the ledger [20] announces further multipliers. For small m both outcomes occur: 143 on F_2 and 88 on F_4 are realized for the tile $(3, 5, 7)$, and 990 on F_3 for the tile $(8, 7, 13)$, the triangle $(169, 286, 315)$, by a tiling of Paliy [21] that is kernel-checked here (`N990.tileCount_990`), and 330 on F_4 for the tile $(7, 8, 13)$, the triangle $(91, 176, 195)$, by an explicit tiling found by exact search and kernel-checked here (`N330.tileCount_330`); the ledger [20] announces this row for every multiplier, and for $a < b$ Paliy's five-piece construction (a scale-c triangle, three scale-a triangles and a $(b - a) \times a$ parallelogram) is proved here for every tile, giving $(2a + b)(a + b)$ as a tile count whenever $0 < a < b$ and $c^2 = a^2 + ab + b^2$ (`F4All.f4_family`); the further rows F3, reversed F4 and F2 of the ledger were rebuilt in exact arithmetic from Paliy's descriptions and checked for every member up to 10^6 (a computation, not formal), as was his tiling of $(961, 1426, 1155)$ by 4830 copies of $(24, 11, 31)$; while the pairs of this type for $N = 33$, 105, 120, 132, 154 and 184 have no tiling, these N not being tile counts. A criterion is not known to us.
- *Type* (T)*, the targets $T_M(e, f)$.* Question 1.12. A tiling exists for $M \geq f$ and none for $M = 1$; for $e = 1$ there is none when $2M \leq f$; for every member there is none when $2Me < f$ (Corollary 1.9); these statements are formal. Open are the scales $M \geq 2$ with $f/(2e) < M < f$, apart from $T_2(1, 3)$, $T_2(2, 3)$ and $T_2(3, 4)$, which have no tiling (Proposition 7.2). In the order of f the first open instances are $T_3(1, 4)$ and, for $e \geq 2$, $T_3(3, 4)$. The exclusions above scale one that are known for infinitely many members all come from a count on one side of the target, and at the open scales that count gives nothing, as far as it has been computed (Section 7.4). For $e = 1$ the open scales reduce to the single scale $M = f - 1$ for each $f \geq 4$ (Proposition 1.11), and two further statements, which are theorems of the Lean library (`SignedDecompE1.signed_decomposition`, `ReframeBWall.reframe_b_wall`, `ReframeC.reframe_c`), delimit the methods. First, $T_M(1, f)$ together with one copy of the tile at scale M attached below its base is the union of three copies of the tile at scale fM; hence no quantity that is additive over the tiles and no invariant with values in a group distinguishes any scale, and an exclusion has to use that every point is covered once. Second, $T_{f-1}(1, f)$ has a tiling if and only if some tiling of $T_f(1, f)$ has the chord at the height of T_1 as a wall, and it has none if every tiling of $T_f(1, f)$ has a wall from the apex along a cevian of length at least $(f - 1)f(f^2 - 1)$; the tilings of $T_f(1, f)$ exist, so this form of the question can be tested. For $e \geq 2$ the first of these two statements holds as well, with the attached copy at scale Me (Proposition 7.4, formal), but the reduction to a single scale does not. The layer of Proposition 7.4 has the height

of $T_e(e,f)$ and relates the scales M and $M+e$; it relates two scales below f only for the members with $(f-e)^2 \geq 2f$. For the members $(f-1,f)$ it exists only at the scales $M \geq f$, so their open scales are not reduced: each scale $2 \leq M < f$ is a separate case, apart from $T_2(2,3)$ and $T_2(3,4)$, and the first open one is $T_3(3,4)$.

- *Type* (T)*, the targets* $W_M(e,f)$. A tiling exists for $M \geq f$ (Theorem 1.5). For the members $(1,f)$ there is none for $M < f$ (Corollary 1.7, formal), and for the members $(f-1,f)$ there is none for $M < f$ (Corollary 1.10, formal); so these members are settled. For the other members there is no tiling when $Me < f$ (Corollary 1.9), none when $M(f-e) \leq e$, and none of $W_2(3,5)$ (Corollary 1.10), all formal. The remaining scales $M < f$ are open; they belong to the members with $2 \leq e \leq f-2$, and the first are $W_3(2,5)$, $W_4(2,5)$, $W_3(3,5)$ and $W_4(3,5)$. By Proposition 7.5 (formal) the open part is, for each of these members, the single target $W_{f-1}(e,f)$, in the direction of exclusion: if $W_{f-1}(e,f)$ has no tiling, then $W_M(e,f)$ has none for any $M < f$, while a tiling of $W_{f-1}(e,f)$ would leave the smaller scales undecided. The first of these targets are $W_4(2,5)$, with 736 tiles, and $W_4(3,5)$, with 656. The question for $W_{f-1}(e,f)$ is equivalent to one about the tilings of $W_f(e,f)$, which exist: whether one of them has the chord that cuts off a translate of $W_{f-1}(e,f)$ at the corner β as a wall.
- *Type* (T)*, the isosceles targets with base angles* $\alpha+\beta$. The triangle with sides tfb, tfb, teb has no tiling when $te < f$ or $(e,f,t) = (k,k+1,2)$ (Proposition 6.4); for the member $(2,3)$ the scale $t=8$ has a tiling, with 320 tiles, kernel-checked (`S5N320.s5_8_2_3_tiling`). The other scales are open.
- *Type* (T)*, the targets with angles* $(\alpha,\alpha,\alpha+2\beta)$ *and* $(2\alpha,\alpha,2\beta)$. The number of tiles is $r^2(f^2-e^2)(2f^2-e^2)$ on the first (`BADL2.theoremA_L2_full`); on the second a layout gives $(2q^2-p^2)(3q^2-p^2)$ tiles for coprime $0<p<q$ with $q^2 > 2p^2$ (`S2G.s2_family_gen`), and a second layout, D. Paliy's, gives the same count for every $0<p<q$ without condition on q^2-2p^2 [21]: five pieces meet at one point, the tile scaled by b twice, by a and by $2q^2-p^2$, and a parallelogram of $2bp^2$ tiles. Its formal proof for all $p<q$ is `S2All.s2_family_all`; every inequality it needs has nonnegative coefficients in $(p, q-p)$. The case $(p,q)=(3,4)$ is the 897-tiling of $(256,368,273)$ by $(12,7,16)$ (`N897.tileCount_897`). With the scaling of tilings, every $k^2(2q^2-p^2)(3q^2-p^2)$, $k \geq 1$, $0<p<q$, is a tile count.

8.3. **Squarefree N.** For squarefree N prime to 3 three of the four odd residue classes modulo 8 are reduced to explicit forms (Theorem 6.7); the class 1 (mod 8) and the multiples of 3 are not treated. The values 455, and 517, 2261, 2485, 3901, are the first of the classes 7 and 5 modulo 8 without a verdict. Computation 6.9 counts the squarefree $N \leq 10^5$ that the present conditions do not decide; some of them are tile counts, for instance 897.

8.4. **Decidability.**

Proposition 8.1 (W)**.** *It is decidable, given N, whether N is a tile count.*

Proof. For fixed N the existence of a triangle $P_1P_2P_3$ and of N triangles $A_iB_iC_i$ that form a dissection of it into congruent triangles is a sentence in the first-order language of ordered fields, with the coordinates of the $3N+3$ points as variables: non-degeneracy is the non-vanishing of determinants; congruence to $A_1B_1C_1$ is the equality of the three squared sides in some order; a point lies in a triangle, or inside it, according to the signs of three determinants; the sentence says that no point lies inside two of the triangles $A_iB_iC_i$ and that a point lies in $P_1P_2P_3$ if and only if it lies in some $A_iB_iC_i$. The sentence is computable from N, and the first-order theory of the real field is decidable [24]. □

This gives no bound that could be used and no description of the set. In Lean explicit criteria exist on two domains only: the primes (Theorem 1.1) and $N \leq 200$ (Theorem 1.3, a conjunction of formal theorems, Section 6.2).

Appendix A. The formalization

A.1. **The definitions.** The library is written in Lean 4 (version 4.30.0) with Mathlib and is in the directory `lean/Erdos634/` of the repository. Its basic definitions are, in mathematical notation:

- `Tri`: three points of the Euclidean plane $\mathbb{R}^2$ that are affinely independent; its carrier is their convex hull.
- `Dissection N`: a target `Tri`, tiles indexed by $\{0, \dots, N-1\}$, the union of the carriers of the tiles equal to the carrier of the target, and the interiors of the carriers of distinct tiles disjoint.
- `Tri.Congruent T U`: there are an isometry g of the plane and a permutation σ of the indices with $g(T_k) = U_{\sigma(k)}$ for the three vertices.
- `CongruentDissection N`: a `Dissection N` with a further `Tri`, the model, to which every tile is congruent.
- `TileCount N`: there is a `CongruentDissection N`.

These are Definition 2.1. They are in the files `Dissection.lean` and `Congruence.lean`, and `TileCount` is in `Draft/TileCountFamilies.lean`. A statement about a target or a tile with given sides is expressed by `HasSides T a b c`: the three sides of T are a, b, c in some order of the vertices.

A.2. **Certificates.** "Machine-checked" has one meaning in this paper: the Lean kernel has checked a proof term of the statement. Two kinds of generated data occur.

A *tiling certificate* is a list of N triangles with coordinates in $\frac{1}{d}\mathbb{Z}[\sqrt{K}]$. The kernel evaluates, in exact arithmetic, that each triangle has the squared sides of the tile, that its vertices lie in the closed target, that for each pair of triangles the line of an edge of one separates them, and that the areas add up to the area of the target. A theorem of the library turns these four facts into a `CongruentDissection`.

A *search certificate* proves that a pair (Δ, T) has no tiling. It is a finite tree. A node is a partial configuration, given by the tiles placed so far, with a convex corner of the uncovered region and one of the two boundary rays there; its children are the six placements of a tile with a vertex at that corner and an edge along that ray, and a child is dead when the tile leaves the target or overlaps a placed tile. That every tiling extending the configuration contains one of these placements is a theorem of the library, also for regions that are not triangles, and so is the soundness of the checker (`Refl` and `ReflS` in `lean/Erdos634/`). The tree is proposed by a floating-point search, which is not trusted: the tree is replayed in exact arithmetic over $\mathbb{Q}(\sqrt{K})$ into Lean terms, and the kernel checks every node. The largest certificate used here, for $N = 135$, has 1 326 256 nodes (Theorem 6.3); the others have at most 345 746. Five certificates of the library use in addition the conjugation $\sqrt{K} \mapsto -\sqrt{K}$ of a tiling, which is Laczkovich's method of conjugate tilings [13]; the pairs that they exclude are also excluded by Corollary 5.3.

A.3. **Aggregates.** The library is built in aggregates that do not import one another, for reasons of memory: `Erdos634.All`, every module whose import closure contains no search certificate, among them the whole proof of Theorem 1.1; `Erdos634.AllSpectrum`, the modules that depend on a search certificate, with the statements on the values up to 200; `Erdos634.AllThreshold`, the results of Section 7; and `AllCert1` to `AllCert5` for the seven largest certificates. These seven are not stored in the repository; they are regenerated from a manifest that records the

search parameters and a hash of each file. In the library, three counting statements that are not used in this paper are proved with `native_decide`; every declaration cited here is recorded with the three standard axioms only. The axioms of a declaration are listed by `#print axioms`; Appendix A.5 says how to rebuild the aggregates and check them.

A.4. **The statements.** Table 3 gives, for each numbered statement, its status (F, W or S, as in Section 1.4) and the Lean declarations, each preceded by its module, the file relative to `lean/Erdos634/`. The last column names the aggregate: A for `All`, Sp for `AllSpectrum`, Th for `AllThreshold`, C for `AllCert1` to `AllCert5`.

Table 3: Numbered statements, their status, and their Lean declarations.

result	status	module: Lean declarations	aggr.
Thm. 1.1	F	`Draft/PrimeCharacterization.lean`: `prime_tile_count_iff`; `Draft/TheoremP6.lean`: `theoremP_6`	A
Thm. 1.2	F	`Draft/BaseBetaFullUnc.lean`: `basebeta_full_unc`; `Draft/UniformThreshold.lean`: `not_betaTilable_one`	A, Th
Thm. 1.3	F	`Draft/Spectrum200g.lean`: `tileCount_iff_le_200g`, `tileCount_iff_le_131`; `Draft/N143.lean`: `tileCount_143`; `Draft/N132Final.lean`: `N132F.not_tileCount_132`; `Draft/Spectrum200Cond4.lean`: `decided_154_156`; `Draft/N161Final.lean`: `N161F.not_tileCount_161`; `Draft/N184Final.lean`: `N184F.not_tileCount_184`; `Draft/N135Final.lean`: `N135F.not_tileCount_135`; `Draft/Spectrum200Excl.lean`: `tileCount_iff_le_200_of_excl`	Sp, C
Thm. 1.4	F	`Draft/ISUMain.lean`: `thmIso_unrestricted`	A
Thm. 1.5	F	`Draft/ResidueSeed.lean`: `Residue.betaTilable_ge`, `Residue.wTilable_ge`	Th
Thm. 1.6	F	`Draft/SideRunRules.lean`: `SideRun.Setup.side_theorem'`	Th
Cor. 1.7	F	`Draft/SideRun.lean`: `SideRun.wTilable_e1_iff`; `Draft/SideRunHalf.lean`: `SideRun.not_betaTilable_e1_le_half`	Th
Thm. 1.8	F	`Draft/SideRunGAngles.lean`: `SideRunG.thm_siderunall_angles_T`, `SideRunG.thm_siderunall_angles_W`	Th
Cor. 1.9	F	`Draft/SideRunGMain.lean`: `SideRunG.ebelow_a`, `SideRunG.ebelow_b`	Th
Cor. 1.10	F	`Draft/SideRunGThick.lean`: `SideRunGA.thickw_a`, `SideRunGA.thickw_b`, `SideRunGA.thickw_c`	Th
Prop. 1.11	F	`Draft/LayerE1.lean`: `LayerE1.betaTilable_succ`, `LayerE1.exists_min_scale`, `LayerE1.threshold_e1_iff_top`	Th
Prop. 2.2	F	`Dissection.lean`: `Dissection.volume_target`; `AngleSumDissection.lean`: `Dissection.hasAngleSums`; `WallEndpoints.lean`: `chain_endpoints`	A
Lem. 2.3	F	`RunPartition.lean`: `run_partition`	A
Prop. 2.4	F	`Ladder.lean`: `ladder`; `Draft/TileCountFamilies.lean`: `tileCount_scale`, `tileCount_two_squares`, `tileCount_three_sq`, `tileCount_six_sq`	A
Prop. 2.5	F	`Draft/LemmaCFinal.lean`: `lemmaC_general`; `Draft/LemmaPProof.lean`: `lemmaP_general`	A
Lem. 2.7	F	`Dissection.lean`: `vertex_multiplicities`; `BaseBetaCorners.lean`: `corner_beta_unique`, `corner_apex_unique`	A
Lem. 3.1	F	`Draft/FluxPaperLemCancel.lean`: `Paper.lem_cancel`	A
Lem. 3.2	F	`Draft/FluxPaper.lean`: `Paper.lem_value_alpha`, `Paper.lem_value_beta`	A
Cor. 3.3	F	`Draft/FluxPaperCorIntAnchored.lean`: `Paper.cor_int_anchored`	A
Lem. 3.4	F	`NonIntegrality.lean`: `k_not_dvd`	A
Prop. 3.5	F	`Rationality.lean`: `tile_rational`	A
Prop. 3.6	F	`Draft/OtherSpectraFull.lean`: `otherspectra`	A
Prop. 3.7	F	`Draft/FDDSpectrumPaper.lean`: `spectrum_paper`	A
Prop. 4.1	F	`Draft/FDBMod12Converse.lean`: `thm_mod12`	A
Lem. 4.3	F	`Draft/PierceScale.lean`: `thm_pierce_scale`; `BaseBetaWalks.lean`: `apex_leftover_nonrepresentable`	A
Prop. 5.1	F	`Draft/FluxCRow5.lean`: `Row5.theoremE_sixty`, `Row5.theoremE_twoPiThree`	A

result	status	module: Lean declarations	aggr.
Thm. 5.2	F	`Draft/E60Count.lean`: `E60Graph.sixty_has_a_relation`; `Draft/E120Count.lean`: `E120Graph.twoPiThree_has_a_relation`; `Draft/E60Irr.lean`: `sixty_commensurable`; `Draft/E120Irr.lean`: `twoPiThree_commensurable`	A, Sp
Cor. 5.3	F	`Draft/E60Irr.lean`: `e60Pair_of_not_isSquare`; `Draft/E120Irr.lean`: `e120Pair_of_not_isSquare`	Sp
Prop. 6.1	F	`Draft/TileCountSpectrum.lean`: `tileCount_seven_sq_iff`, `tileCount_eleven_sq_iff`; `Draft/SporadicCounts.lean`: `tc77`; `R84TilingBridge.lean`, `R88TilingBridge.lean`, `R126TilingBridge.lean`, `R189TilingBridge.lean`; `Draft/N143.lean`: `dissection_143_unit`	A, Sp
Prop. 6.2	F	`Draft/N135Red.lean`: `N135.tileCount_135_iff`	Sp
Thm. 6.3	F	`Draft/N135Final.lean`: `N135F.not_tileCount_135`	C
Prop. 6.4	F	`Draft/S5Family.lean`: `NSQM.s5_base_word2_hasSides`, `NSQM.s5_family_2k_hasSides`; `Draft/AtomShape.lean`: `NSQM.atom_shape_te`, `NSQM.atom_not_squarefree`; `Draft/B3Prime.lean`: `NSQM.b3prime`	A
Prop. 6.5	F	`Draft/GAUFinal.lean`: `GAU.gapA_sq`	A
Prop. 6.6	F	`Draft/TwoPrime.lean`: `two_prime_tileCount_iff`	Sp
Thm. 6.7	F	`Draft/Sqfree3Mod8.lean`: `not_tile_sqfree_3mod8`; `Draft/Sqfree7Mod8.lean`: `tile_sqfree_7mod8_form`, `sector_below_455`; `Draft/QPExact.lean`: `tile_sqfree_5mod8_explicit`, `sector5_below_517`	Sp
Comp. 6.9	S	none	
Thm. 7.1	F	`Draft/WSeedGAssemble.lean`: `WSeedG.beta_seed`; `Draft/WSeedGW.lean`: `WSeedG.w_seed`	Th
Prop. 7.2	F	`Draft/BaseBetaScale2.lean`: `basebeta_scale_two`, `scale_two_1_2_exists`	C
Prop. 7.3	F	`Draft/SideRunGCornerMainB.lean`: `SideRunGA.thm_cornerrun`, `SideRunGA.thm_cornerrun_addendum`	Th
Prop. 7.4	F	`Draft/SignedDecompW.lean`: `signed_decomposition_member_W`; `Draft/LayerG.lean`: `layer_real`, `layer_c_i`; `Draft/LayerGChord.lean`: `layer_c_ii`	Th
Prop. 7.5	F	`Draft/WStrip.lean`: `WStripG.wstrip_step`; `Draft/WStripChord.lean`: `wstrip_c`, `wstrip_d`	Th
Prop. 7.6	F	`Draft/ResidueSeed.lean`: `Residue.threshold_iff_small`; `Draft/UniformThreshold.lean`: `threshold_1_2`; `Draft/UniformThresholdCert.lean`: `threshold_1_3`, `threshold_2_3`, `threshold_1_4_of_ne_three`, `threshold_3_4_of_ne_three`; `Draft/E1ScaleTwo.lean`: `thresholdAt_e1_of`; `Draft/SideRunHalf.lean`: `SideRun.thresholdAt_e1_le_half`	Th, C
Prop. 7.7	F	`Draft/ThresholdBlind.lean`: `Blind.invariant_blind`	Th
Prop. 8.1	W	none	

Three qualifications apply to the table. First, Theorem 1.3 is a conjunction of formal theorems, as described in Section 6.2. Second, no formal statement of the table has yet been compared with the mathematical one by an independent reader (Section 1.4). Third, the formal statements of the lemmas of Sections 2 and 3 are phrased for the structures of the library; for instance Lemma 2.3 is stated with a linear functional for the line and two auxiliary points for the clamping, and Lemma 2.7 as statements on the multiplicities.

A.5. **Checking the formal proofs.** The library builds with Lean 4 version 4.30.0 and Mathlib at the tag `v4.30.0`, as pinned in `lean/lakefile.toml` and `lean/lean-toolchain`. A reader checks a numbered statement by building the aggregate named in Table 3 and printing the axioms of its declarations. The aggregates are to be built one at a time: a single `lean` process needs up to about 13.5 GB of memory for the largest modules. The aggregate `All` contains the whole proof of Theorem 1.1 and no search certificate. The seven certificates used by `AllCert1` to `AllCert5` are not stored in the repository; the script `code/regen_cert.sh` regenerates each of them from the manifest `data/certificates/MANIFEST.tsv`, by rerunning the search and the exact replay, and compares the SHA-256 hashes of the search tree and of the generated Lean

files with those of the manifest. The kernel check of the largest of them, for $N = 135$, takes several hours.

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